Dobrushin uniqueness for the four-dimensional SU(2) Wilson lattice gauge measure for $\beta < 2/9$, with interval/Lean certificates

Version 1.4

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28 October 2026 (v1.4: literature update, §7; v1.3 of 28 September 2026, not publicly released)

Abstract

We write out Dobrushin's uniqueness criterion, in its Wasserstein (W_1) form, for the $SU(2)$ Wilson lattice gauge measure on $\mathbb{Z}^4$, and compute the single-link influence coefficient *exactly*, uniformly in the field, for two metrics on $SU(2) \cong S^3$. For the Euclidean (chordal) metric of $\mathbb{R}^4$, the metric used by Levin (1980), we show that the one-site coefficient in a field of strength $\kappa \le 4/3$ equals $\mathbb{E}_\kappa[x_\perp^2]$, the second moment of a coordinate orthogonal to the field; it is maximal at zero field, where it equals $1/4$. Hence each of the 18 neighbouring links has influence exactly $\beta/4$, the Dobrushin sum is $9\beta/2$, and the (DLR) Gibbs measure exists and is unique for $\beta < 2/9 \approx 0.2222$, in the normalisation $\beta \sum_p (1 - \frac{1}{2} \operatorname{Re} \operatorname{Tr} U_p)$. This confirms, for $N = 4$ and $|g| \le 4/3$, the maximum-at-zero-field property conjectured by Levin (Remark 3.7), which Bhattacharjee and Bhattacharya prove for $N = 4$ and all fields [17, Theorem 7]; accordingly, $2/9$ is exactly Levin's published bound transferred to the Wilson link, $2\sqrt{5}/45 \approx 0.0994$, with his factor $\sqrt{5}$ replaced by 1. The proof bounds the covariance by the L^1 norm of an explicit divergence-free vector field on the ball with prescribed normal flux, eliminates the direction analytically, and reduces the claim to a one-variable inequality on $[0, 4/3]$, proved by interval arithmetic (two independent implementations); its final rational step is checked in Lean 4. For the geodesic metric (the main result of v1.2 of this note) the exact coefficient is $8\beta/(9\pi)$, uniformly for $\kappa \le 3\pi/8$, which gives the weaker range $\beta < \pi/16 \approx 0.196$ by the same method. Each threshold is sharp for its criterion. The single-site machinery is not new: Levin (1980, Euclidean metric) and Royer (1986, geodesic metric) treat rotator models, and Levin's many-body version already covers the Wilson model; none of the cited papers mentions gauge theories. For comparison, the explicit range of Shen–Zhu–Zhu is $\beta < 1/12$, the total-variation criterion (Gross 1979, Simon 1979) gives $\beta < 1/18$ transferred, and global Bakry–Émery on link variables cannot pass $\beta = 1/8$. No statement on decay of correlations, spectral gap or log-Sobolev inequality is claimed. *Prior public statement.* The uniqueness range $\beta < 2/9$ for the four-dimensional $SU(2)$ model, with exponential decay and a lattice mass gap, and the all-field bound $W_1(\nu_b, \nu_{b'}) \le \frac{1}{4}|b - b'|$ on S^3, were stated publicly, with proofs, in the preprint [17] of 18 October 2026 (Theorems E, F and 7 there). This note was prepared independently and makes *no priority claim.* Its own contributions are the exact identification of the one-link chordal coefficient as $\mathbb{E}_\kappa[x_\perp^2]$ for $\kappa \le 4/3$, with the explicit remainder $\frac{1}{4} - \mathbb{E}_\kappa[x_\perp^2] \ge \kappa^2/(96\,\bar{Z}(\kappa))$, the exact geodesic coefficient $8\beta/(9\pi)$ for $\kappa \le 3\pi/8$, two independent interval-arithmetic certificates, Lean-checked final steps, and an explicit transfer of the published constants of Levin, Royer and Shen–Zhu–Zhu with their conventions.

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Status labels. Every claim carries one label: [proved] (complete argument written here or in the cited source); [interval-certified] (rigorous modulo trust in the interval library and in the reductions of §§4.2–4.3 and §5.1, which are proved but not formalised); [Lean-checked] (kernel-checked in Lean 4, axioms `propext`, `Classical.choice`, `Quot.sound` only; this label never covers more than the Lean statements quoted verbatim in §4.4 and §5.2, which are rational inequalities, conditional implications and threshold arithmetic); [read] (statement read in the original source, location given); [conjecture]. The label [numerical] marks floating-point evidence which is *not* part of any proof.

1 Statement and conventions

Let E be the set of (positively oriented) links of $\mathbb{Z}^4$ and $\mathcal{P}$ the set of unoriented plaquettes. A configuration is $U = (U_\ell)_{\ell\in E} \in SU(2)^E$; U_p is the ordered product around p (its real trace does not depend on the starting point or orientation, since $\operatorname{Re}\operatorname{Tr} W = \operatorname{Re}\operatorname{Tr} W^{-1}$ on $SU(2)$). For a finite $\Lambda \subset E$ and boundary condition $\bar{U}$, the Wilson specification is

$$\mu_\Lambda^{\bar{U}}(dU_\Lambda) \;\propto\; \exp\Big(-\beta \sum_{p\cap\Lambda\neq\emptyset} \big(1 - \tfrac{1}{2}\operatorname{Re}\operatorname{Tr} U_p\big)\Big) \prod_{\ell\in\Lambda} dU_\ell, \tag{1}$$

with dU the normalised Haar measure. Throughout, $\beta = \beta_W$ is the *Wilson* coupling, and the bare coupling g is *defined* by $\beta_W = 2N/g^2$, i.e. $\beta_W = 4/g^2$ for $SU(2)$; the coupling g_0 of Osterwalder–Seiler is a different normalisation (Table 1). A Gibbs measure is a probability measure on $SU(2)^E$ satisfying the DLR equations for (1). We identify $SU(2)$ with the unit sphere $S^3 \subset \mathbb{R}^4$ (unit quaternions) and use two metrics on it: the *chordal* (Euclidean) distance $\rho_{\mathrm{ch}}(x,y) = |x-y|$ of $\mathbb{R}^4$, of diameter 2, and the *geodesic* (great-circle) distance ρ, of diameter π. They generate the same topology and Borel σ-algebra.

convention	weight	relation to β_W (for $SU(2)$)
Wilson (this note)	$\exp(-\beta_W \sum_p (1 - \frac{1}{N}\operatorname{Re}\operatorname{Tr} U_p))$	—
$\frac{\beta}{N}\operatorname{Re}\operatorname{Tr}$	$\exp(\frac{\beta}{N}\sum_p \operatorname{Re}\operatorname{Tr} U_p)$	$\beta_W = \beta$
raw trace	$\exp(\beta \sum_p \operatorname{Re}\operatorname{Tr} U_p)$	$\beta_W = 2\beta$
't Hooft (SZZ, CNS)	$\exp(N\beta \sum_p \operatorname{Re}\operatorname{Tr} U_p)$	$\beta_W = N^2\beta = 4\beta$
Osterwalder–Seiler	$\exp(\frac{1}{2g_0^2}\sum_P \operatorname{Re}\chi(g_P))$	$\beta_W = 1/g_0^2$ (hypothesis*), i.e. $g_0 = g/2$

Table 1: Conventions. The 't Hooft line is [2, (1.2)] (see §7); the Osterwalder–Seiler line is [3, (2.1)–(2.2), p. 442], read on the page image ([read]). *The conversion $\beta_W = 1/g_0^2$ is a *hypothesis*: it holds if χ is the fundamental character and each geometric plaquette is counted once in $\sum_P$. On p. 442, χ is "a character on $\mathcal{G}$" and plaquettes are "closed loops $uvxy$"; neither condition is stated there. The conversion is used only to translate thresholds, never in a proof.

Theorem 1 (Main result)**.** *For $0 \le \beta < 2/9 \approx 0.2222$ there is exactly one Gibbs measure for the $SU(2)$ Wilson specification (1) on $\mathbb{Z}^4$. Equivalently: $g^2 > 18$ with g defined by $\beta_W = 2N/g^2 = 4/g^2$; in the 't Hooft normalisation of [2], $\beta_{\mathrm{tH}} < 1/18 \approx 0.0556$; in the normalisation of [3] (under the hypothesis of Table 1), $g_0^2 > 9/2$.*

Status: existence [proved] (Proposition 3); uniqueness [read] (Dobrushin 1970, Thm 4) + [proved] (§3, chordal metric) + Theorem 6, which is [proved] + [interval-certified]; its last rational step and the threshold arithmetic are [Lean-checked] (§4.4).

Theorem 2 (Geodesic criterion; the main result of v1.2)**.** *With the geodesic metric ρ, Dobrushin's sum for (1) equals $16\beta/\pi$ for $0 \le \beta < \pi/16 \approx 0.19635$; so the geodesic W_1 criterion gives uniqueness for $\beta < \pi/16$ and fails at $\beta = \pi/16$.*

Status: [proved] (§3) + Theorem 10, which is [proved] for $\kappa = 0$ and [interval-certified] for $\kappa > 0$, with its last rational step [Lean-checked] (§5.2). The uniqueness statement for $\beta < \pi/16$ is contained in Theorem 1; what Theorem 2 adds is the exact geodesic coefficient $8\beta/(9\pi)$ and its field-uniformity.

Each threshold is *sharp for its criterion*: with single-site conditionals, the chordal sum equals $9\beta/2$ and the geodesic sum $16\beta/\pi$ exactly (Proposition 5), so the chordal criterion fails at $\beta = 2/9$ and the geodesic one at $\beta = \pi/16$. Nothing is claimed about other metrics, block or multi-site versions of the criterion, or the true uniqueness region.

2 Dobrushin's theorem and its hypotheses

We use the original Russian article [1] (Math-Net.Ru scan `tvp1856`; the quotations below were read on the rendered page images; English wording is ours).

Setting (§1, p. 469). T is a countable set; the random variables ξ_t take values in a measurable space $(X, \mathfrak{B})$. Dobrushin's standing assumption is that all probability measures on finite products of $(X, \mathfrak{B})$ are perfect ("все вероятностные меры на прямых произведениях конечного числа экземпляров измеримого пространства $(X, \mathfrak{B})$ совершенны"), and for this it suffices (verbatim) that "$\mathfrak{B}$ было σ-алгеброй всех борелевских подмножеств для некоторой метрики, превращающей X в полное сепарабельное метрическое пространство". Given an integer function ν on T,

$$G_{t_0} = \{t : \nu(t) \le \nu(t_0),\ t \ne t_0,\ t \in T\}, \qquad t_0 \in T \qquad [1,\ (1.2)],$$

and a field has conditional distributions $\{p_{t_0}\}$ if $\mathbf{P}\{\xi_{t_0} \in B \mid \xi_t, t \in G_{t_0}\} = p_{t_0}(B \mid x_t, t \in G_{t_0})$ [1, (1.3)].

Wasserstein distance (§3, p. 479). "Назовем *расстоянием Васерштейна* (см. [12]) число $R(P, Q) = \inf \mathrm{M}\rho(\eta, \zeta)$, (3.1) где нижняя грань берется по всем парам случайных величин $\eta, \zeta \ldots$ таким, что η имеет распределение P и ζ — распределение Q". This is W_1 defined by couplings, for the metric ρ of X (any metric making X complete and separable).

Theorem 4 (§5, p. 488), verbatim. "Для того чтобы существовало не более, чем одно, случайное поле $\{\xi_t, t \in T\}$ с заданными условными распределениями $\{p_{t_0}(\cdot|\cdot), t_0 \in T\}$, для которого при некоторых $x_0 \in X$ и $C < \infty$ $\mathrm{M}\rho(\xi_t, x_0) \le C < \infty$, $t \in T$, (5.1) достаточно, чтобы при некотором $d < 1$ и всех $t_0 \in T$ $\sum_{t \in G_{t_0}} r(t, t_0) \le d < 1$, (5.2) где числа $r(t, t_0) \ldots$ таковы, что расстояние Васерштейна $R(p_{t_0}(\cdot|x^1_t, t \in G_{t_0}), p_{t_0}(\cdot|x^2_t, t \in G_{t_0})) \le \sum_{t \in G_{t_0}} r(t, t_0)\rho(x^1_t, x^2_t)$ (5.3) $\ldots$ Условие (5.3) выполнено, в частности, если $r(\bar{t}, t_0) = \sup R(\cdots)/\rho(x^1_{\bar{t}}, x^2_{\bar{t}})$, (5.4) где верхняя грань берется по всем парам условий $\ldots$, отличным лишь при $t = \bar{t}$." (Translation: at most one field with the given conditionals and property (5.1) exists provided that, for some $d < 1$ and all t_0, (5.2) holds with numbers r satisfying the Wasserstein bound (5.3); (5.3) holds in particular for the supremum (5.4) over pairs of conditions differing only at $\bar{t}$.)

Theorem 1 (existence, pp. 472–473). Two conditions suffice; on p. 473 Dobrushin writes "Условие 1 заведомо выполнено, если X — бикомпакт. (Можно положить $h(x) \equiv 1$.)" and, in the Markov case (conditionals depending on finitely many x_t, $t \in U_{t_0}$), "условие 2 означает непрерывность $p_{t_0}(\cdot \mid x_t, t \in U_{t_0})$ по переменным x_t в топологии слабой сходимости".

Verification of the hypotheses for (1). Let d be either ρ_{ch} or ρ.

1. $T = E$ is countable; $X = SU(2) \cong S^3$ with d is compact, hence complete and separable; $\mathfrak{B}$ is its Borel σ-algebra (the same for both metrics). [proved]

2. Take $\nu \equiv 0$; then $G_{t_0} = T \setminus \{t_0\}$ (the case Dobrushin himself mentions on p. 488, "В случае $G_{t_0} = \{t : t \neq t_0\}$"). The kernel $p_\ell(\cdot \mid U_{E\setminus\ell})$ is the single-link conditional law (2) below; it has a density continuous in all variables, hence is jointly measurable. Every DLR measure for (1) satisfies [1, (1.3)] with these kernels. So uniqueness in Dobrushin's class implies uniqueness of the Gibbs measure. [proved]

3. (5.1) holds for every field with $x_0 = \mathbf{1}$ and $C = \operatorname{diam} S^3$ (2 for ρ_{ch}, π for ρ). [proved]

4. (5.2)–(5.4): p_ℓ depends only on the 18 neighbouring links (§3), so $r(t,\ell) = 0$ for all other t. With r the supremum (5.4): for $d = \rho_{\mathrm{ch}}$ and $\beta < 2/9$, $\sum_t r(t,\ell) = 9\beta/2$ (Proposition 5 and Theorem 6); take $d = 9\beta/2 < 1$. For $d = \rho$ and $\beta < \pi/16$, $\sum_t r(t,\ell) = 16\beta/\pi$ (Theorem 10). [proved] given Theorems 6 and 10.

5. Existence: Theorem 1 applies (Condition 1 because X is compact; Condition 2 because the conditional density is continuous in the neighbouring links), but it yields a field with the prescribed *single-site* conditionals. We do not use it: existence of a measure satisfying the DLR equations for *every* finite Λ is proved directly in Proposition 3. [proved]

A second modern statement of the W_1 criterion on complete metric single-spin spaces is [8, Thm 3.5] (p. 8 of arXiv v4) [read].

Proposition 3 (Existence in the DLR sense)**.** *For every $\beta \geq 0$ there is at least one probability measure μ on $SU(2)^E$ such that $\mu\gamma_\Delta = \mu$ for every finite $\Delta \subset E$, where $\gamma_\Delta(\cdot \mid \bar U) = \mu_\Delta^{\bar U} \otimes \delta_{\bar U_{E\setminus\Delta}}$ is the kernel (1).*

Status: [proved].

Proof. Write $H_\Delta(U) = \sum_{p\cap\Delta\neq\emptyset}(1 - \frac{1}{2}\operatorname{Re}\operatorname{Tr} U_p)$, a finite sum of continuous functions of finitely many links, with $0 \leq H_\Delta \leq h_\Delta := 2\,|\{p : p\cap\Delta \neq \emptyset\}|$. Thus

$$\gamma_\Delta f(\bar U) = Z_\Delta(\bar U)^{-1} \int f(U_\Delta, \bar U_{E\setminus\Delta})\, e^{-\beta H_\Delta(U_\Delta, \bar U)}\, dU_\Delta, \qquad Z_\Delta \geq e^{-\beta h_\Delta} > 0.$$

(a) Feller property. If f is continuous on the compact space $SU(2)^E$ (product topology), the integrand is jointly continuous in $(U_\Delta, \bar U)$ and bounded, so $\gamma_\Delta f$ and Z_Δ are continuous by dominated convergence, and $\gamma_\Delta f$ is continuous. *(b) Consistency.* For finite $\Delta \subset \Lambda$ one has $\gamma_\Lambda\gamma_\Delta = \gamma_\Lambda$: the density of $\gamma_\Lambda(\cdot \mid \bar U)$ is $e^{-\beta H_\Lambda}/Z_\Lambda$ and $H_\Lambda = H_\Delta + H'$ with H' independent of U_Δ (the plaquettes meeting Λ but not Δ), so by Fubini integrating $\gamma_\Delta f$ against γ_Λ first in U_Δ reproduces $\gamma_\Lambda f$. *(c) Limit.* Fix $\bar U$ and finite $\Lambda_n \uparrow E$, and let $\mu_n = \gamma_{\Lambda_n}(\cdot \mid \bar U)$. By compactness (Prokhorov, or Riesz–Markov on $C(SU(2)^E)$) a subsequence converges weakly to some μ. For finite Δ and continuous f, (b) gives $\mu_n(\gamma_\Delta f) = \mu_n(f)$ as soon as $\Delta \subset \Lambda_n$; by (a) both sides converge along the subsequence, so $\mu(\gamma_\Delta f) = \mu(f)$ for all continuous f, hence $\mu\gamma_\Delta = \mu$. □

Every such μ satisfies in particular [1, (1.3)] with $G_{t_0} = T \setminus \{t_0\}$ (take $\Delta = \{\ell\}$), which is the direction used for uniqueness in item 2.

3 The single-link conditional law and the influence matrix

Quaternion form. For unit quaternions u, w, $\frac{1}{2}\operatorname{Re}\operatorname{Tr}(uw) = \langle u, \bar{w}\rangle_{\mathbb{R}^4}$. Link ℓ lies in $2(d-1) = 6$ plaquettes; for each such p, $\frac{1}{2}\operatorname{Re}\operatorname{Tr} U_p = \frac{1}{2}\operatorname{Re}\operatorname{Tr}(U_\ell W_p)$ where the *staple* W_p is a product of the three other links of p (or of their inverses; we use $\operatorname{Re}\operatorname{Tr} A = \operatorname{Re}\operatorname{Tr} A^{-1}$ to put U_ℓ first). Hence

$$p_\ell(du \mid \cdot) = q_s(du) \propto e^{\beta\langle u,s\rangle}\, du, \qquad s = \sum_{p\ni\ell} \overline{W_p} \in \mathbb{R}^4, \quad |s| \le 6. \tag{2}$$

Write $\kappa = \beta|s| \in [0, 6\beta]$, $\hat{s} = s/|s|$, and $q_\kappa \propto e^{\kappa\langle u,\hat{s}\rangle}$. In the notation of Royer [10, p. 31], $q_s = \nu(\beta s)$ on S^n with $n = 3$: "For each vector $h \in \mathbb{R}^{n+1}$ let $\nu(h)$ be the probability measure on S^n, the density of which is proportional to $x \to \exp(x.h)$" [read], on the Numdam page image; in Levin's notation [11, (3.11), p. 68] it is $\mu_{\mathbf{h}}$ with $\mathbf{h} = \beta s$ and $N = 4$ [read]. The single-site analysis below is therefore an instance of the setting of Levin [11] and Royer [10] for n-vector (rotator) models; what is specific to the gauge model is the staple structure and the count of neighbours. In fact the whole Wilson model is a 4-vector model with four-body interactions in the sense of Levin [11, Remark 3.6, (3.48), p. 73]: writing $u_1, \dots, u_4 \in S^3 \subset \mathbb{R}^4$ for the links of p (with $u^{-1} = \bar{u}$), $\frac{1}{2}\operatorname{Re}\operatorname{Tr} U_p = \langle u_1 u_2 \bar{u}_3 \bar{u}_4, \mathbf{1}\rangle$ is a multilinear function of (u_1, u_2, u_3, u_4). [proved]

Why 18 neighbours. Each of the 6 plaquettes through ℓ contains 3 further links. Two distinct plaquettes through ℓ span different coordinate planes, or the same plane on opposite sides of ℓ, so they share no link other than ℓ. Hence the $6 \times 3 = 18$ links are pairwise distinct, each shares exactly one plaquette with ℓ, and a change of one of them changes exactly one staple. In general dimension the count is $3 \cdot 2(d-1)$. [proved]

Definition 4. For $\kappa \ge 0$ and a unit vector $v \in \mathbb{R}^4$ put

$$D_{\mathrm{ch}}(\kappa, v) = \sup\{\operatorname{Cov}_{q_\kappa}(f, \langle u, v\rangle) : f \text{ 1-Lipschitz for } \rho_{\mathrm{ch}}\},$$
$$D(\kappa, v) = \sup\{\operatorname{Cov}_{q_\kappa}(f, \langle u, v\rangle) : f \text{ 1-Lipschitz for } \rho\},$$

and $D_{\mathrm{ch}}(\kappa) = \sup_{|v|=1} D_{\mathrm{ch}}(\kappa, v)$, $D(\kappa) = \sup_{|v|=1} D(\kappa, v)$. By rotation invariance about $\hat{s}$, both depend on v only through the angle $\alpha \in [0, \pi]$ between v and $\hat{s}$; replacing f by $-f$ shows that they bound $|\operatorname{Cov}|$.

Proposition 5 (Influence coefficients)**.** *Let $r_{\mathrm{ch}}(y, \ell)$ and $r(y, \ell)$ be Dobrushin's coefficient (5.4) for a neighbour y of ℓ, for the metrics ρ_{ch} and ρ respectively. Then*

$$\beta\, D_{\mathrm{ch}}(0) \;\le\; r_{\mathrm{ch}}(y, \ell) \;\le\; \beta \sup_{0\le\kappa\le 6\beta} D_{\mathrm{ch}}(\kappa), \qquad \beta\, D(0) \;\le\; r(y, \ell) \;\le\; \beta \sup_{0\le\kappa\le 6\beta} D(\kappa).$$

With $D_{\mathrm{ch}}(0) = 1/4$ and Theorem 6, for $\beta < 2/9$: $r_{\mathrm{ch}}(y, \ell) = \beta/4$ and $\sum_y r_{\mathrm{ch}}(y, \ell) = 18\beta/4 = 9\beta/2$. With $D(0) = 8/(9\pi)$ (Lemma 11) and Theorem 10, for $\beta < \pi/16$: $r(y, \ell) = 8\beta/(9\pi)$ and $\sum_y r(y, \ell) = 16\beta/\pi$.

Status: [proved] (the upper bounds use Kantorovich–Rubinstein duality [8, Thm 10.8], quoted below). The geodesic upper bound is the argument of Royer [10, (3)–(4), p. 31] for $\nu(h)$ on S^n, here with $n = 3$, $K = \beta$ and z the moving staple: verbatim, "$\mathrm{R}(\nu(Kz_1 + \zeta), \nu(Kz_2 + \zeta)) \leqslant |K|(C(n)/(n+1))^{1/2} r(z_1, z_2)$. (3)" and, along a minimal geodesic $z(t)$, "$\left|\frac{d}{dt}\int f(y)\nu(Kz(t) + \zeta, dy)\right| = |K \operatorname{cov}(f(y), y.z'(t))|$ (4)" [read]. The chordal upper bound is the argument of Levin, who writes the distance "as the variation of an expectation along an appropriate path in **h**-space" [11, p. 68], the straight segment [11, (3.35), p. 71] [read]. Royer and Levin then bound the covariance by field-uniform constants; we instead compute $\sup_f$ of the covariance exactly, as a function of $\kappa = |h|$.

Proof. Upper bounds. Let U, U' differ only at y, and let $W = AU_y^{\pm 1}B$ be the unique staple containing y (A, B fixed). The map $U_y \mapsto \overline{AU_y^{\pm 1}B}$ is a composition of left and right multiplications by unit quaternions and of quaternion conjugation, hence the restriction to S^3 of an orthogonal map of $\mathbb{R}^4$; it is an isometry for both ρ_{ch} and ρ. *Chordal:* let $s(t) = s + t(s' - s)$, $t \in [0,1]$, the straight segment between the two values of s; then $|s(t)| \le 6$ by convexity and $|s' - s| = |\overline{W'} - \overline{W}| = \rho_{\mathrm{ch}}(U_y, U_y')$. For f 1-Lipschitz for ρ_{ch},

$$\int f \, d(q_{s'} - q_s) = \int_0^1 \beta \, \mathrm{Cov}_{q_{s(t)}} \left(f, \langle u, s' - s\rangle\right) dt \le \beta \, \rho_{\mathrm{ch}}(U_y, U_y') \sup_{\kappa \le 6\beta} D_{\mathrm{ch}}(\kappa).$$

Geodesic: along a constant-speed geodesic $t \mapsto U_y(t)$, $t \in [0,1]$, from U_y to U_y', the vector $s(t)$ moves on a great circle of the unit sphere translated by the other five staples, with $|\dot{s}(t)| = \rho(U_y, U_y')$, and $|s(t)| \le 6$. For f 1-Lipschitz for ρ,

$$\int f \, d(q_{s(1)} - q_{s(0)}) = \int_0^1 \beta \, \mathrm{Cov}_{q_{s(t)}} \left(f, \langle u, \dot{s}(t)\rangle\right) dt \le \beta \, \rho(U_y, U_y') \sup_{\kappa \le 6\beta} D(\kappa).$$

In both cases the supremum over 1-Lipschitz f of the left side is Dobrushin's Wasserstein distance (3.1) for the corresponding metric, by the Kantorovich–Rubinstein duality as stated in [8, Thm 10.8, p. 53] (verbatim): "Let μ and ν be two probability measures on a metric space $(\mathcal{X}, \rho)$ such that $\int_{\mathcal{X}} \rho(x, x_0)\, \mu(dx) < +\infty$ and $\int_{\mathcal{X}} \rho(x, x_0)\, \nu(dx) < +\infty$, for some $x_0 \in \mathcal{X}$. Then the Wasserstein metric equals the Kantorovich-Rubinstein metric, i.e., $\rho_W(\mu, \nu) = \sup_{f \in \mathrm{Lip}_1(\mathcal{X})} \left(\int_{\mathcal{X}} f(x)\, \mu(dx) - \int_{\mathcal{X}} f(x)\, \nu(dx)\right)$" [read], where ρ_W is the infimum over couplings [8, (2.1), p. 6], i.e. Dobrushin's (3.1). The moment condition is trivial on the compact S^3. Dividing by the distance of U_y and U_y' and taking the supremum gives the bounds on (5.4).

Lower bounds. The 18 neighbours are distinct and each staple involves three of them only, so the six staples can be prescribed independently. Given a unit $v \in \mathbb{R}^4$, choose the staple containing y so that $w := \overline{W} \perp v$, and the other five staples (unit vectors) with sum $-w$ (possible since $|-w| = 1 \le 5$); then $s = 0$. Moving U_y along a unit-speed geodesic moves $\overline{W}$ isometrically along a geodesic of S^3 through w; choosing it with initial velocity v (tangent at w since $v \perp w$) gives $\overline{W}(\varepsilon) = \cos\varepsilon\, w + \sin\varepsilon\, v$, so $s(\varepsilon) = \varepsilon v + O(\varepsilon^2)$, $\rho(U_y, U_y(\varepsilon)) = \varepsilon$ and $\rho_{\mathrm{ch}}(U_y, U_y(\varepsilon)) = 2\sin(\varepsilon/2) = \varepsilon + O(\varepsilon^3)$. For fixed 1-Lipschitz f (for either metric), $W_1(q_0, q_{s(\varepsilon)}) \ge \int f\, d(q_{s(\varepsilon)} - q_0) = \beta\varepsilon \, \mathrm{Cov}_{q_0}(f, \langle u, v\rangle) + O(\varepsilon^2)$ (only the trivial direction of the duality is used). Dividing by the distance and letting $\varepsilon \to 0$, then optimising f, gives $r_{\mathrm{ch}}(y, \ell) \ge \beta D_{\mathrm{ch}}(0, v)$ and $r(y, \ell) \ge \beta D(0, v)$; both zero-field values are independent of v (Lemmas 7 and 11). □

4 The chordal one-link identity

Throughout this section $\hat{s} = e_1$, $x_1 = \langle u, e_1\rangle$, $x_2 = \langle u, e_2\rangle$ with $e_2 \perp e_1$, $\bar{\sigma}$ is the normalised Haar measure of S^3, and $dq_\kappa = e^{\kappa x_1}\, d\bar{\sigma}/\bar{Z}(\kappa)$ with

$$\bar{Z}(\kappa) = \mathbb{E}_{\bar{\sigma}} e^{\kappa x_1} = \sum_{k \ge 0} \frac{\kappa^{2k}}{4^k\, k!\, (k+1)!} = \frac{2I_1(\kappa)}{\kappa}$$

(the marginal law of x_1 under $\bar{\sigma}$ is $\frac{2}{\pi}\sqrt{1 - t^2}\, dt$). We write $\mathbb{E}_\kappa$ for $\mathbb{E}_{q_\kappa}$ and $m_1 = \mathbb{E}_\kappa x_1 = \bar{Z}'/\bar{Z}$.

Theorem 6 (Chordal one-link identity)**.** *For all $\kappa \in [0, 4/3]$,*

$$D_{\mathrm{ch}}(\kappa) = D_{\mathrm{ch}}(\kappa, e_2) = \mathbb{E}_\kappa[x_2^2] = \frac{1}{3}\Big(1 - \frac{\bar{Z}''(\kappa)}{\bar{Z}(\kappa)}\Big),$$

$$\frac{1}{4} - D_{\mathrm{ch}}(\kappa) = \frac{4\bar{Z}''(\kappa) - \bar{Z}(\kappa)}{12\,\bar{Z}(\kappa)} \;\geq\; \frac{\kappa^2}{96\,\bar{Z}(\kappa)} \;\geq\; 0.0084\,\kappa^2.$$

In particular $D_{\mathrm{ch}}(\kappa) \leq 1/4$, *with equality only at* $\kappa = 0$, *and* $D_{\mathrm{ch}}(\kappa) = \frac{1}{4} - \frac{\kappa^2}{96} + \frac{\kappa^4}{1536} - \frac{\kappa^6}{23040} + \frac{13\kappa^8}{4423680} - \cdots$. *Since* $6 \cdot 2/9 = 4/3$, *this covers* $\kappa \leq 6\beta$ *for all* $\beta < 2/9$.

Status: $\kappa = 0$ and the lower bound $D_{\mathrm{ch}}(\kappa) \geq \mathbb{E}_\kappa[x_2^2]$ [proved] (Lemma 7); the upper bound [proved] (§§4.2–4.3) up to the one-variable inequality (5), which is [interval-certified] (§4.4; its last rational step is [Lean-checked]); the identity for $\frac{1}{4} - D_{\mathrm{ch}}$ and the bound $\kappa^2/(96\bar{Z})$ [proved]; the constant 0.0084 [interval-certified]; the expansion [proved] (exact rational arithmetic). The inequality $6\beta < 4/3$ for $\beta < 2/9$ is [Lean-checked] (`kappa_max_lt`).

4.1 Reflection: the value at zero field and the lower bound

Lemma 7 (Reflection)**.** *Let* v *be a unit vector,* $R_v x = x - 2\langle x, v\rangle v$, *and* q *a probability measure on* S^3 *with* $R_{v\#}q = q$. *Then* $\sup\{\mathrm{Cov}_q(f, \langle x, v\rangle) : f$ *1-Lipschitz for* $\rho_{\mathrm{ch}}\} = \mathbb{E}_q\langle x, v\rangle^2$. *In particular* $D_{\mathrm{ch}}(0, v) = \mathbb{E}_{\bar{\sigma}}\langle x, v\rangle^2 = 1/4$ *for every* v, *and* $D_{\mathrm{ch}}(\kappa, e_2) = \mathbb{E}_\kappa[x_2^2]$ *for every* $\kappa \geq 0$. *[proved]*

Proof. Put $u = \langle x, v\rangle$; $u \circ R_v = -u$, so $\mathbb{E}_q u = 0$ and $\mathrm{Cov}_q(f, u) = \mathbb{E}_q[fu] = \frac{1}{2}\mathbb{E}_q[(f(x) - f(R_v x))u]$. Since $|x - R_v x| = 2|u|$, this is $\leq \mathbb{E}_q u^2$, with equality for $f = u$, which is 1-Lipschitz. For $q = \bar{\sigma}$, $\mathbb{E}u^2 = \frac{1}{4}\mathbb{E}|x|^2 = \frac{1}{4}$; for $q = q_\kappa$ and $v = e_2$, R_{e_2} preserves x_1, hence q_κ. □

The case $v \perp \mathbf{h}$ of Lemma 7 is Levin's Lemma 3.2: for $\hat{\mathbf{e}} \perp \mathbf{h}$, "$|\mu_{\mathbf{h}}[(\mathbf{x} \cdot \hat{\mathbf{e}})f]| \leqslant \frac{1}{N}$ (3.12)", proved by the same reflection "$\leqslant \mu_{\mathbf{h}}(x_1^2)$ (3.14)" [11, p. 68] [read]. The content of Theorem 6 is that, for $N = 4$ and $\kappa \leq 4/3$, no other direction does better.

4.2 Divergence-free fields

Lemma 8 (Flux bound)**.** *Let* $u = \langle x, v\rangle$ *and* $w = (u - \mathbb{E}_\kappa u)\, e^{\kappa x_1}/\bar{Z}(\kappa)$ *on* S^3, *so that* $\mathrm{Cov}_\kappa(f, u) = \mathbb{E}_{\bar{\sigma}}[fw]$. *Let* $V \in C^1(\bar{B}; \mathbb{R}^4)$ *on the closed unit ball* $\bar{B} \subset \mathbb{R}^4$ *with* $\operatorname{div} V = 0$ *in* B *and* $V(x) \cdot x = w(x)$ *on* S^3. *Then, for every* f *1-Lipschitz for* ρ_{ch},

$$\mathrm{Cov}_\kappa(f, u) \leq \frac{1}{2\pi^2} \int_B |V(y)|\, dy.$$

[proved]

Proof. McShane's extension $F(y) = \min_{x \in S^3}(f(x) + |y - x|)$ is 1-Lipschitz on $\mathbb{R}^4$ and equals f on S^3 (here the chordal metric is essential: it is the restriction of the Euclidean one); by Rademacher's theorem $|\nabla F| \leq 1$ a.e. Since FV is Lipschitz on $\bar{B}$ and $\operatorname{div}(FV) = \nabla F \cdot V$ a.e., the divergence theorem and $|S^3| = 2\pi^2$ give $\mathbb{E}_{\bar{\sigma}}[fw] = \frac{1}{2\pi^2}\int_{S^3} F\, V \cdot x\, d\sigma = \frac{1}{2\pi^2}\int_B \nabla F \cdot V\, dy \leq \frac{1}{2\pi^2}\int_B |V|\, dy$. □

For $\kappa = 0$ the constant field $V = v$ is admissible and gives $|B|/|S^3| = (\pi^2/2)/(2\pi^2) = 1/4$: a second proof of $D_{\mathrm{ch}}(0) \leq 1/4$. Applied to $F(y) = \langle v, y\rangle$ (equality case), the same computation gives $\frac{1}{2\pi^2}\int_B \langle v, V\rangle\, dy = \mathbb{E}_{\bar{\sigma}}[uw] = \mathrm{Var}_\kappa(u)$ for every admissible V.

The explicit field. Write $y = (y_1, z)$ with $z = (0, y_2, y_3, y_4)$, $R = \sqrt{1 - y_1^2}$, $E(t) = e^{\kappa t}/\bar{Z}(\kappa)$, and let A be the bounded solution on $[-1, 1]$ of

$$tA(t) - \tfrac{1}{3}(1 - t^2)A'(t) = q(t) := (t - m_1)\, e^{\kappa t}/\bar{Z}(\kappa), \qquad A(t) = \frac{3}{(1 - t^2)^{3/2}} \int_t^1 q(s)\sqrt{1 - s^2}\, ds \tag{3}$$

(for $t<0$ equivalently $-\int_{-1}^{t}$, since $\int_{-1}^{1} q\sqrt{1-s^2}\,ds \propto \mathbb{E}_\kappa x_1 - m_1 = 0$). For $v = \cos\alpha\, e_1 + \sin\alpha\, e_2$ put

$$V = \cos\alpha\,[A(y_1)\,e_1 - \tfrac{1}{3}A'(y_1)\,z] + \sin\alpha\,E(y_1)\,e_2. \tag{4}$$

Then $\operatorname{div} V = \cos\alpha\,(A' - A') + \sin\alpha\,\partial_{y_2}E(y_1) = 0$ (z has three components), and on S^3, where $|z|^2 = 1 - x_1^2$, $V\cdot x = \cos\alpha\, q(x_1) + \sin\alpha\, x_2 E(x_1) = (u - \cos\alpha\, m_1)e^{\kappa x_1}/\bar{Z} = w$, because $\mathbb{E}_\kappa u = \cos\alpha\, m_1$. [proved] *Series and regularity.* $A = \sum_{j\ge 0}\kappa^j A_j$ where A_j is the unique polynomial solution of $L[A_j] = q_j$, $L[p] = tp - \frac{1}{3}(1-t^2)p'$, q_j the coefficient of κ^j in q: L maps degree d to degree $d+1$ with leading factor $1 + d/3$, the homogeneous solutions $c(1-t^2)^{-3/2}$ are unbounded, and the solvability condition $\int q_j\sqrt{1-t^2} = 0$ holds order by order; so A_j has degree j ($A_0 = 1$, $A_1 = \frac{3}{4}t$, $A_2 = \frac{3}{10}t^2 - \frac{7}{40}$). A is holomorphic in κ for $|\kappa| \le 2$ (where $|\bar{Z}| \ge 2 - \bar{Z}(2) > 0.4$) and bounded there, so Cauchy's estimates and Markov's inequality $\|p'\|_\infty \le (\deg p)^2\|p\|_\infty$ make the series and its first two t-derivatives converge uniformly on $[-1,1]$ for $\kappa \le 4/3$; hence $V \in C^1(\bar{B})$ [proved]. The explicit bounds are those of the certificate (§4.4).

4.3 Elimination of the direction

Proposition 9. *Let $m(y_1) = \min(A, E) - |A'|R/6$ and assume $m > 0$ on $[-1,1]$. With*

$$J_0 = \frac{2}{5\pi}\int_{-1}^{1}\Big(\frac{A'}{3}\Big)^2\frac{R^5}{2m}\,dy_1, \qquad J_1' = \frac{2}{3\pi}\int_{-1}^{1}(E-A)^2\frac{R^3}{2m}\,dy_1, \qquad J_1 = J_1' + \tfrac{2}{3}J_0,$$

one has, for all α, $D_{\mathrm{ch}}(\kappa,\alpha) \le \cos^2\alpha\,(\operatorname{Var}_\kappa x_1 + \max(J_0, J_1)) + \sin^2\alpha\,\mathbb{E}_\kappa x_2^2$. [proved]

Proof. By Lemma 8 applied to (4), $D_{\mathrm{ch}}(\kappa,\alpha) \le \frac{1}{2\pi^2}\int_B |V|$. Pointwise, $\langle v, V\rangle = \cos^2\alpha\, A + \sin^2\alpha\, E - \cos\alpha\sin\alpha\,\frac{A'}{3}y_2 \ge m(y_1) > 0$ (as $|\cos\alpha\sin\alpha| \le \frac{1}{2}$ and $|y_2| \le R$), and $\sqrt{a^2+b^2} \le a + b^2/(2a)$ for $a > 0$ gives $|V| \le \langle v, V\rangle + |V_\perp|^2/(2m)$, where, in the orthonormal frame (v, n, e_3, e_4) with $n = -\sin\alpha\, e_1 + \cos\alpha\, e_2$, $|V_\perp|^2 = \cos^2\alpha\,[(\sin\alpha(E-A) - \cos\alpha\frac{A'}{3}y_2)^2 + (\frac{A'}{3})^2(y_3^2+y_4^2)]$. The integral of $\langle v, V\rangle$ is $2\pi^2\operatorname{Var}_\kappa(u) = 2\pi^2(\cos^2\alpha\operatorname{Var}_\kappa x_1 + \sin^2\alpha\,\mathbb{E}_\kappa x_2^2)$ (end of §4.2; $\operatorname{Cov}_\kappa(x_1,x_2) = 0$ and $\mathbb{E}_\kappa x_2 = 0$ by the symmetry $x_2 \mapsto -x_2$). On the section $\{|z| \le R\}$ of the ball, the cross term is odd in y_2, and $\int 1 = \frac{4}{3}\pi R^3$, $\int y_2^2 = \frac{4}{15}\pi R^5$, $\int(y_3^2+y_4^2) = \frac{8}{15}\pi R^5$. Dividing by $2\pi^2$, with $s = \sin^2\alpha$, the excess is at most $\cos^2\alpha\,[sJ_1' + (1 - \frac{s}{3})J_0]$, affine in $s \in [0,1]$, hence at most $\cos^2\alpha\,\max(J_0, J_1)$. □

Reduction to one variable. If, for a given κ, $m > 0$ on $[-1,1]$ and $\operatorname{Var}_\kappa x_1 + \max(J_0, J_1) \le \mathbb{E}_\kappa x_2^2$, Proposition 9 gives $D_{\mathrm{ch}}(\kappa,\alpha) \le \mathbb{E}_\kappa x_2^2$ for all α (the convex combination `etape_alpha`, [Lean-checked]), and Lemma 7 gives equality at $\alpha = \pi/2$. Now $N := \bar{Z}^2(\mathbb{E}_\kappa x_2^2 - \operatorname{Var}_\kappa x_1) = \frac{1}{3}\bar{Z}^2 - \frac{4}{3}\bar{Z}''\bar{Z} + \bar{Z}'^2 = \kappa^2\tilde{N}(\kappa)$ with $\tilde{N}$ a power series, and $A' = \kappa\tilde{A}$, $E - A = \kappa\tilde{D}$ with power series $\tilde{A}, \tilde{D}$ (since $A_0 = E_0 = 1$). Writing $\tilde{J}_0, \tilde{J}_1$ for J_0, J_1 with $A', E-A$ replaced by $\tilde{A}, \tilde{D}$ (so $J_i = \kappa^2\tilde{J}_i$), it suffices to prove

$$\frac{\tilde{N}(\kappa)}{\bar{Z}(\kappa)^2} > \max\big(\tilde{J}_0(\kappa), \tilde{J}_1(\kappa)\big) \quad\text{and}\quad m > 0 \text{ on } [-1,1], \qquad \text{for all } \kappa \in [0, 4/3] \tag{5}$$

(the step from (5) to the inequality before it is `etape_kappa2`, [Lean-checked]). At $\kappa = 0$, exactly, $\tilde{N}/\bar{Z}^2 = 1/48$, $\tilde{A} = \frac{3}{4}$, $\tilde{D} = \frac{t}{4}$, $m = 1$, and $\tilde{J}_0 = \tilde{J}_1 = 1/256$: the margin is $13/768$, so (5) is *not* degenerate at zero field, unlike the geodesic inequality of §5. [proved]

The bound by 1/4. $\mathbb{E}_\kappa x_2^2 = \frac{1}{3}(1 - \mathbb{E}_\kappa x_1^2)$ and $\mathbb{E}_\kappa x_1^2 = \bar{Z}''/\bar{Z}$, so $\frac{1}{4} - \mathbb{E}_\kappa x_2^2 = (4\bar{Z}'' - \bar{Z})/(12\bar{Z})$. The coefficient of κ^{2k} in $4\bar{Z}'' - \bar{Z}$ is $\frac{3k}{k+2}\cdot\frac{1}{4^k k!(k+1)!} \ge 0$ (from $z_{2k+2}/z_{2k} = \frac{1}{4(k+1)(k+2)}$), equal to $\frac{1}{8}$ for $k = 1$. Hence $\frac{1}{4} - \mathbb{E}_\kappa x_2^2 \ge \kappa^2/(96\bar{Z}(\kappa)) \ge \kappa^2/(96\bar{Z}(4/3))$, and $\bar{Z}(4/3) \in [1.2393065416564178344 \pm 3\cdot 10^{-20}]$ gives $1/(96\bar{Z}(4/3)) \ge 0.0084052$ [interval-certified]. The expansion of Theorem 6 follows by exact series division. [proved]

4.4 Computer-assisted part

Interval evaluation. Script `certificate_chordal/cert_ch.py`, Arb ball arithmetic (python-flint 0.9.0), working precision 128 bits and series length cap 200, both mandatory arguments, set explicitly and read back. All series in κ ($\bar{Z}, \bar{Z}', \bar{Z}'', 1/\bar{Z}$, m_1, $m_1/\bar{Z}$, $\bar{Z}^2$, $\bar{Z}''\bar{Z}$, $\bar{Z}'^2$, N) are computed to length $L = D + 5 = 85$ ($D = 80$); guard 1 refuses cap $< L$, guard 2 asserts `.prec` $= L$ on each of these ten series and products (trailing zeros are padded explicitly, since `coeffs()` drops them). The A_j are obtained by exact downward recursion; the solvability residual contains 0 for all $j \le 80$. *Tails:* all orders > 80 are bounded, not neglected: Cauchy's estimates on $|\kappa| \le 2$ ($|\bar{Z}| \ge 0.409363$, $|m_1| \le 1.6830$, $|E| \le 18.050$, $|A| \le 228.21$, using $\int_t^1 \sqrt{1-s^2}\,ds \le \frac{\pi}{2}(1-t^2)^{3/2}$) and on $|\kappa| \le 4$ for N, and Markov's inequality for A_j'. At $\kappa = 4/3$ the tails are $1.9 \cdot 10^{-8}$ ($\tilde{A}$), $3.0 \cdot 10^{-12}$ ($\tilde{D}$) and $7.3 \cdot 10^{-39}$ (N). *Boxes:* $[0, 4/3]$ is cut into 256 boxes $[i/192, (i+1)/192]$, κ being a ball in each; the series are folded into polynomials in $t = y_1$, and $[-1, 1]$ is cut into 1024 dyadic sub-intervals, on which the polynomials are evaluated in centred form $p(t_c) + p'(T)(T - t_c)$. On each sub-interval $m > 0$ is *asserted*, and $\tilde{J}_0$, $\tilde{J}_1$ are bounded above by upper Riemann sums (with exact upper bounds for R^3, R^5). $\tilde{N}$ is bounded below on the box (minus its tail), and $\tilde{Z}$ above by its value at the right end (nonnegative coefficients).

Result. All 256 boxes pass; the worst margin is $1.142 \cdot 10^{-2}$, on the last box $[85/64, 4/3]$ (lower bound $1.6883 \cdot 10^{-2}$ for $\tilde{N}/\bar{Z}^2$ against $\tilde{J}_1 \le 5.463 \cdot 10^{-3}$, $\tilde{J}_0 \le 3.975 \cdot 10^{-3}$, $m \ge 0.192$); at $\kappa \in [0, 1/192]$ the margin is $1.690 \cdot 10^{-2}$. The right-hand side can be multiplied by about 3 before the inequality fails. [interval-certified]

Gates. All logs return code 0 unless a refusal is expected (logs in the archive).

gate	outcome
cap $\in \{85, 100, 200, 300\}$	identical results in all 256 boxes; cap 10 and $84 = L - 1$ refused by guard 1, and by guard 2 when guard 1 is disabled
order $D \in \{40, 60, 80, 100\}$	the bound moves ($\tilde{J}_0$ of the last box $4.8604 \cdot 10^{-3}$, $3.97576 \cdot 10^{-3}$, $3.975228508 \cdot 10^{-3}$, $3.975228233 \cdot 10^{-3}$) and stabilises; all four runs pass
sub-intervals $\in \{256, \ldots, 2048\}$	upper Riemann sums strictly decreasing, step halving ($O(1/N_T)$)
precision 128 vs 192	$\max \lvert\Delta\tilde{J}_1\rvert = 6.7 \cdot 10^{-14}$; all boxes pass
boxes $\in \{64, 128, 512\}$	all pass; worst margin 1.106, 1.130, $1.148 \cdot 10^{-2}$
exact gate (fractions, no Arb)	solvability residuals exactly 0 for $j \le 80$; $\tilde{N}_0 = 1/48$, odd orders 0; all 6891 exact coefficients lie in the Arb balls
numerical gate ([numerical])	series A vs direct quadrature $1.3 \cdot 10^{-15}$; $\lvert\operatorname{div} V\rvert \le 4 \cdot 10^{-11}$; $\lvert V \cdot x - w\rvert \le 7 \cdot 10^{-16}$; full flux bound $\frac{1}{2\pi^2}\int_B \lvert V\rvert$ by 3D quadrature within $1.7 \cdot 10^{-4}$ of the linear programs of `calc/lp2d_m14.log` (e.g. 0.20892, 0.22140, 0.23332 vs 0.20887, 0.22145, 0.23324 at $\kappa = 4/3$, $\alpha = 0, \pi/4, \pi/2$)

Independent check. A second implementation (`independent_check_chordal/`), written without reading or importing the certificate code, recomputes everything with different tools: exact rational coefficients up to order 100 (own downward solver; solvability residuals exactly 0 and $L[A_j] = q_j$ checked exactly), `mpmath.iv` interval arithmetic instead of Arb, elementary tail bounds instead of complex Cauchy estimates ($\lvert[\kappa^n]1/\bar{Z}\rvert \le 2^{-n}$ by induction, and $(1 - t^2)^{-3/2}\int_t^1 \sqrt{1-s^2}\,ds \le 1$ on $[0, 1)$ instead of $\pi/2$), a uniform partition in θ ($t = \cos\theta$) with exact weights $\int \sin^4$, $\int \sin^6$, and naive Horner evaluation. On its own partition (200 boxes of width 1/150) it certifies (5) on all of $[0, 4/3]$ (worst margin $1.1395 \cdot 10^{-2}$, $m \ge 0.188$); it also recertifies the 256 boxes of the Lean data file and checks, two-sidedly, that none of

the exported rationals contradicts its own enclosures. Its negative gates fail as they should (precision sabotaged to 53 bits: refused; series shorter than $D+2$: refused; left side of (5) scaled by 0.30: fails on the last box, while 0.35 passes). Exact checks: $\mathbb{E}_{\bar\sigma} u^2 = 1/4$ and the cost $1/4$ of the reflection coupling (SymPy), the coefficient $3k/(k+2)$ symbolically for all k, $c_2 = 1/96$; a Monte-Carlo evaluation of the full flux bound at $\kappa = 4/3$ (0.20884, 0.22131, 0.23322) agrees with the linear programs [numerical]. [interval-certified]

Lean. Commit `1b0f039c53c1e2a11c496cc5a1cb7774ab5a010e` of the author's Lean library (parent `7a211ed`, which holds the geodesic modules of §5.2; three new files, no existing file changed), module `KevYM/Certif/DobrushinCordale/`. The three modules are reproduced byte for byte in the standalone project `lean/` of the accompanying archive. Lean toolchain `v4.34.0`. Main statements (verbatim):

```
theorem sound (L : List Boite) (a c : ℚ) (hch : chaine L a c = true) (hok : L.all Boite.ok = true)
    (Nt J0 J1 : ℝ → ℝ)
    (hN : ∀ b ∈ L, ∀ κ : ℝ, (b.k0 : ℝ) ≤ κ → κ ≤ (b.k1 : ℝ) → (b.lb : ℝ) ≤ Nt κ)
    (hJ0 : ∀ b ∈ L, ∀ κ : ℝ, (b.k0 : ℝ) ≤ κ → κ ≤ (b.k1 : ℝ) → J0 κ ≤ (b.h0 : ℝ))
    (hJ1 : ∀ b ∈ L, ∀ κ : ℝ, (b.k0 : ℝ) ≤ κ → κ ≤ (b.k1 : ℝ) → J1 κ ≤ (b.h1 : ℝ))
    (κ : ℝ) (ha : (a : ℝ) ≤ κ) (hc : κ ≤ (c : ℝ)) : max (J0 κ) (J1 κ) < Nt κ
theorem etape_kappa2 (k Nt Jt V1 E2 : ℝ) (hE : E2 - V1 = k ^ 2 * Nt) (h : Jt < Nt) :
    V1 + k ^ 2 * Jt ≤ E2
theorem etape_alpha (V1 E2 J c s : ℝ) (hcs : c ^ 2 + s ^ 2 = 1) (h : V1 + J ≤ E2) :
    c ^ 2 * (V1 + J) + s ^ 2 * E2 ≤ E2
theorem somme_dobrushin_lt_one (β : ℝ) (hβ : β < 2 / 9) : 18 * β * (1 / 4) < 1
theorem kappa_max_lt (β : ℝ) (hβ : β < 2 / 9) : 6 * β < 4 / 3
theorem pi_div_16_lt : Real.pi / 16 < 2 / 9
theorem certificat_cordal (Nt J0 J1 : ℝ → ℝ)
    (hN : ∀ b ∈ boites, ∀ κ : ℝ, (b.k0 : ℝ) ≤ κ → κ ≤ (b.k1 : ℝ) → (b.lb : ℝ) ≤ Nt κ)
    (hJ0 : ∀ b ∈ boites, ∀ κ : ℝ, (b.k0 : ℝ) ≤ κ → κ ≤ (b.k1 : ℝ) → J0 κ ≤ (b.h0 : ℝ))
    (hJ1 : ∀ b ∈ boites, ∀ κ : ℝ, (b.k0 : ℝ) ≤ κ → κ ≤ (b.k1 : ℝ) → J1 κ ≤ (b.h1 : ℝ))
    (κ : ℝ) (h0 : 0 ≤ κ) (hK : κ ≤ 4 / 3) : max (J0 κ) (J1 κ) < Nt κ
```

Here `Boite.ok b` is `h0 < lb && h1 < lb && k0 ≤ k1`, `chaine` checks that the boxes are chained from a to c, and `boites` is the list of the 256 exported boxes $\langle k_0, k_1, lb, h_0, h_1\rangle$ (Arb bounds rounded outward to the grid 2^{-40}, re-checked exactly before export); `boites_ok` and `boites_chaine` are proved by `decide +kernel`. The file also proves $18 \cdot (2/9) \cdot \frac{1}{4} = 1$, the converse for $\beta \geq 2/9$, and $6 \cdot (2/9) = 4/3$. Build clean (0 warnings); all 16 theorems of the module depend only on `propext`, `Classical.choice`, `Quot.sound`; the rebuild from `git archive 1b0f039` succeeds, and the standalone project `lean/` of the archive builds from an empty `.lake` with the same axiom report (`lean/logs/build_from_zero.log`). [Lean-checked]

What Lean covers, exactly. `certificat_cordal` is a conditional implication: *if* real functions `Nt`, `J0`, `J1` (standing for $\tilde N/\bar Z^2$, $\tilde J_0$, $\tilde J_1$) are bounded box by box by the exported rationals, *then* $\max(\tilde J_0, \tilde J_1) < \tilde N/\bar Z^2$ on $[0, 4/3]$. Lean also checks the two algebraic steps `etape_alpha`, `etape_kappa2` and the threshold arithmetic. Not in Lean: the flux bound, the explicit field and Proposition 9 (proved on paper, §§4.2–4.3), the bounds of each box and the positivity of m ([interval-certified]), and the constant 0.0084.

Artefacts (SHA-256). Paths are relative to the accompanying archive.

file	SHA-256
`certificate_chordal/cert_ch.py`	`d83aa23b2fd20707b08076c87053aa8c9bc9e277ca15d78be95e5afa7dab14c0`
`certificate_chordal/export_lean.py`	`f9f59cbeb0a0ac0c609ce7471b97ea3c69ea7f7330ac4c5975c152416b374ac4`
`certificate_chordal/cert_D80_NT1024_NK256_cap200_p128.json`	`aa5da0f05c1ee230782c30bd451981df2100ca75a9fc46ca73a6b7e54c765bb5`
`lean/KevYM/Certif/DobrushinCordale/Donnees.lean` (= `1b0f039`)	`72a0901472475b3a6d98033692b899e32c5f9b2a1daab7b9ed5c22416dc462fe`

5 The geodesic one-link bound

This section is the main result of v1.2 of this note, unchanged in substance. It gives the exact field-uniform coefficient for the geodesic metric, hence Theorem 2.

Theorem 10 (One-link bound, geodesic metric)**.** *For all $\kappa \in [0, 59/50]$ and all unit v,*

$$D(\kappa, v) \;\le\; \frac{8}{9\pi} - 0.0085\,\kappa^2,$$

with equality at $\kappa = 0$. Since $6 \cdot \pi/16 = 3\pi/8 < 59/50$, this covers $\kappa \le 6\beta$ for all $\beta < \pi/16$.

Status: $\kappa = 0$ [proved]; $\kappa > 0$ [interval-certified]. Precisely: Lean proves only that the monotone bound `bP` is < 0 in each of the 160 boxes, i.e. $\Phi < 0$ on $[0, K^2]$ *if* the exported rationals bound the coefficients ([Lean-checked], §5.2); the constant 0.0085 and the covering of $\alpha \in [0, \pi]$ by the boxes (box $\mapsto$ α-interval) are [interval-certified], not in Lean. The inequality $3\pi/8 < 59/50$ is [Lean-checked] (`kappa_max_lt`).

Lemma 11 ($\kappa = 0$)**.** *$D(0, v) = 8/(9\pi)$ for every unit v. [proved]*

Proof. Let $\gamma \in [0, \pi]$ be the angle between u and v; under Haar measure its law is $\frac{2}{\pi}\sin^2\gamma\, d\gamma$. *Upper bound:* $\cos\gamma$ is a first eigenfunction of $-\Delta_{S^3}$ (eigenvalue 3), so $\psi = \cos\gamma/3$ solves $-\Delta\psi = \langle u, v\rangle$ and, for 1-Lipschitz f, $\mathrm{Cov}(f, \langle u, v\rangle) = \mathbb{E}[\nabla f \cdot \nabla\psi] \le \mathbb{E}|\nabla\psi| = \frac{1}{3} \cdot \frac{2}{\pi}\int_0^\pi \sin^3\gamma\, d\gamma = \frac{8}{9\pi}$. *Lower bound:* $f = \pi/2 - \gamma$ is 1-Lipschitz and (integration by parts)

$$\mathrm{Cov}(f, \cos\gamma) = -\mathbb{E}[\gamma\cos\gamma] = -\frac{2}{\pi}\int_0^\pi \gamma\cos\gamma\sin^2\gamma\, d\gamma = -\frac{2}{\pi}\Big(-\frac{4}{9}\Big) = \frac{8}{9\pi}.$$

□

5.1 Reduction (proved, not formalised)

Step 1: transport. Let $L = \Delta_{S^3} + \kappa\nabla x \cdot \nabla$ with $x = \langle u, \hat{s}\rangle$; L is symmetric in $L^2(q_\kappa)$, with Dirichlet form $\mathbb{E}_{q_\kappa}[\nabla\varphi\cdot\nabla\psi]$. *Weak solutions (Lax–Milgram).* Let H be the space of $\varphi \in H^1(S^3)$ with $\mathbb{E}_{q_\kappa}\varphi = 0$, with the inner product $\mathbb{E}_{q_\kappa}[\nabla\varphi \cdot \nabla\psi]$; it is a Hilbert space by the Poincaré inequality of Step 3 ($\lambda > 0$). For $g \in L^2(q_\kappa)$ with $\mathbb{E}_{q_\kappa} g = 0$, the functional $\varphi \mapsto \mathbb{E}_{q_\kappa}[\varphi g]$ is bounded on H by $\|g\|_2\lambda^{-1/2}\|\nabla\varphi\|_2$, so by Riesz there is a unique $\psi \in H$ with $\mathbb{E}[\nabla\varphi\cdot\nabla\psi] = \mathbb{E}[\varphi g]$ for all $\varphi \in H^1$ (constants integrate against g to zero); we write $-L\psi = g$ in this weak sense, which is all that is used below. With $g = \langle u, v\rangle - \mathbb{E}_{q_\kappa}\langle u, v\rangle$ and f 1-Lipschitz for ρ (so $f \in H^1$ with $|\nabla f| \le 1$ a.e., by Rademacher's theorem), $\mathrm{Cov}_{q_\kappa}(f, \langle u, v\rangle) = \mathbb{E}[\nabla f \cdot \nabla\psi_v] \le \mathbb{E}|\nabla\psi_v|$, so $D(\kappa, v) \le \mathbb{E}_{q_\kappa}|\nabla\psi_v|$. [proved]

Step 2: reduced ODEs and exact series. Write $v = \cos\alpha\,\hat{s} + \sin\alpha\, e$ with $e \perp \hat{s}$, $y = \langle u, e\rangle$, and seek $\psi_v = \cos\alpha\, g(x) + \sin\alpha\, y\, h(x)$, $p = g'$. Using $\Delta_{S^3} g(x) = (1-x^2)g'' - 3xg'$, $\Delta_{S^3}(yh) = y[(1-x^2)h'' - 5xh' - 3h]$, $|\nabla x|^2 = 1 - x^2$ and $\nabla x \cdot \nabla y = -xy$, the equation reduces to

$$(1-x^2)(p' + \kappa p) - 3xp = -(x - m), \qquad (1-x^2)(h'' + \kappa h') - 5xh' - (3 + \kappa x)h = -1,$$

with $m = \mathbb{E}_{q_\kappa} x$. Expanding $p = \sum_n \kappa^n p_n$, $h = \sum_n \kappa^n h_n$, $m = \sum_n \kappa^n m_n$ gives polynomials p_n, h_n with rational coefficients, computed exactly; p_n, h_n have parity $(-1)^n$ and $m_n = 0$ for n even (asserted in exact arithmetic by the certificate script). For the truncation ψ_N at order N, the residual is $L\psi_N + (\langle u, v\rangle - \cos\alpha\, m) = \cos\alpha\,(m_N - m) + R'$ with $R' = \kappa^{N+1}\big[\cos\alpha(1 - x^2)p_N + \sin\alpha\, y((1-x^2)h_N' - xh_N)\big]$ (identity also checked symbolically in $\mathbb{R}^4$ at rational points of S^3). Since the full residual R has q_κ-mean zero, $|\cos\alpha(m_N - m)| \le \|R'\|_\infty$, hence $\|R\|_\infty \le 2\kappa^{N+1}\rho_N$, where ρ_N is the sum of ℓ^1 norms of the coefficients of the two bracketed polynomials ($\rho_{16} \approx 8.1 \cdot 10^{-10}$). [proved]

Step 3: correction via Poincaré. Let $-L\varphi = R$; then $\psi_v = \psi_N + \varphi$ exactly. The density of q_κ w.r.t. Haar measure has log-oscillation at most 2κ and the spectral gap of $-\Delta_{S^3}$ is 3, so by the Holley–Stroock perturbation argument [9] the Poincaré constant of q_κ satisfies $\lambda \geq 3e^{-2\kappa}$. (Proof: write $dq_\kappa = h\, d\sigma$ with σ the Haar measure, $h = e^{\kappa x}/\tilde{Z}$, so $e^{-\kappa}/\tilde{Z} \leq h \leq e^{\kappa}/\tilde{Z}$; then $\mathrm{Var}_q f \leq \mathbb{E}_q(f - \mathbb{E}_\sigma f)^2 \leq \frac{e^\kappa}{\tilde{Z}} \mathrm{Var}_\sigma f \leq \frac{e^\kappa}{3\tilde{Z}} \mathbb{E}_\sigma |\nabla f|^2 \leq \frac{e^{2\kappa}}{3} \mathbb{E}_q |\nabla f|^2$.) Then $\mathbb{E}|\nabla\varphi|^2 = \mathbb{E}[\varphi R] \leq \|R\|_2 (\mathbb{E}|\nabla\varphi|^2/\lambda)^{1/2}$, whence

$$\mathbb{E}|\nabla\varphi| \leq \|R\|_2/\sqrt{\lambda} \leq \varepsilon_N(\kappa) := 2\kappa^{N+1}\rho_N e^\kappa/\sqrt{3}.$$

So $D(\kappa, v) \leq \mathbb{E}|\nabla\psi_N| + \varepsilon_N(\kappa)$. [proved]

Step 4: AM–GM. With $\varrho = \sin\gamma/3 = |\nabla\psi_0|$ ($\cos\gamma = \langle u, v\rangle$), pointwise $|X| \leq (|X|^2/\varrho + \varrho)/2$, hence $\mathbb{E}|\nabla\psi_N| \leq U_N := \mathbb{E}[(|\nabla\psi_N|^2/\varrho + \varrho)/2]$, with equality at $\kappa = 0$. [proved]

Step 5: polynomial coordinates. The image of the uniform measure on S^3 under $u \mapsto (\langle u, \hat{s}\rangle, \langle u, e\rangle)$ is uniform on the unit disc, and $|\nabla\psi_N|^2$ is a polynomial in (x, y, κ). In the rotated frame $s = \langle u, v\rangle$, $t = \sqrt{1 - s^2}\,\tau$, the singular weight $1/\varrho$ cancels against the Jacobian and

$$U_N(\kappa, \alpha)\, Z(\kappa) = \iint_{[-1,1]^2} \left[\tfrac{3}{2}|\nabla\psi_N|^2 + \tfrac{1-s^2}{6}\right] e^{\kappa x}\, ds\, d\tau, \qquad Z(\kappa) = \pi \sum_{k\geq 0} \frac{(\kappa^2/4)^k}{k!\,(k+1)!} = \pi \bar{Z}(\kappa).$$

At $\kappa = 0$ the integral is $8/9$ and $Z(0) = \pi$, recovering $8/(9\pi)$. The coefficient of κ^j is the integral of a polynomial of degree $\leq j + 2$ in (s, τ): the terms odd in τ integrate to zero (also exactly on symmetric Gauss nodes), and in the remaining terms $\sqrt{1 - s^2}$ occurs only squared. [proved]

Step 6: exact factorisation of κ^2. Put $\Delta(\kappa, \alpha) := (U_N + \varepsilon_N - \frac{8}{9\pi})Z = \sum_j E_j(\alpha)\kappa^j +$ (remainders). $E_0 = 0$ exactly ($I_0 = 8/9$), and $E_j = 0$ for j odd by the parity $u \mapsto -u$, i.e. $(s, \tau) \mapsto (-s, -\tau)$. Hence $\Delta = \kappa^2\Phi(w)$ with $w = \kappa^2$ and $\Phi(w) = \sum_{k\geq 1} E_{2k} w^{k-1} +$ (Taylor remainder $+ \varepsilon_N Z)/\kappa^2$; the last two terms are increasing in κ and are bounded at $\kappa = K = 59/50$. This is what makes the comparison at $\kappa \to 0$ non-degenerate: no two touching balls are ever compared. [proved]

Since Z is increasing, $\Phi \leq -c_*$ on $[0, K^2]$ implies $D(\kappa, v) \leq \frac{8}{9\pi} - \frac{c_*}{Z(K)}\kappa^2$.

Remark 12 (second order, exact). An independent exact computation (SymPy, harmonic components in $\mathbb{R}^4$; code not included in the archive) gives $E_2(\alpha) = -(1309\cos^2\alpha + 779)/21600$, maximal at $\alpha = \pi/2$. Locally as $\kappa \to 0$ the method cannot give a coefficient better than $779/(21600\pi) \approx 0.0115$ in front of κ^2; along the route actually used ($c_*/Z(K)$, uniform on $[0, K]$), since $c_* \leq -\sup_\alpha \Phi(0) = 779/21600$, the ceiling is $779/(21600\, Z(K)) \approx 0.0097$. The certified 0.0085 is below both. (Status: exact computation; the same E_2 and E_4 are reproduced in exact rationals, by a different method, by the independent check `independent_check_geodesic/`.)

5.2 Computer-assisted part

Interval evaluation. Parameters: series order $N = 16$; Gauss–Legendre with $N_G = 22$ nodes per direction, exact for polynomial degree $\leq 2N_G - 1 = 43$, hence for all $j \leq D = 41$; $\alpha \in [0, \pi]$ cut into 160 boxes of width $\pi/160$, $(\cos\alpha, \sin\alpha)$ enclosed by balls; Arb ball arithmetic (python-flint 0.9.0) with working precision 128 bits and series length cap 200, both set explicitly and read back. Nodes and weights are rigorous balls (`arb.legendre_p_root`). Orders $\kappa^0, \ldots, \kappa^{41}$ are computed; the orders ≥ 42 of $e^{\kappa x}$ are covered by the Lagrange bound $|\mathrm{rem}|/\kappa^2 \leq 4\kappa^{D-1}e^\kappa \sum_i \sup|A_i|/(D + 1 - i)! \leq 1.53 \cdot 10^{-21}$ at $\kappa = K$. On each box, $\Phi(w) \leq \sum_k \max(c_k \cdot 0^k, c_k K^{2k})$ with c_k upper bounds of E_{2k+2} (the constant term absorbing the remainder bounds).

Result. All 160 boxes pass in one piece in w; the worst bound is $\Phi \le -0.03175361$ (box 75, α near $\pi/2$); $Z(59/50) \in [3.721046472 \pm 4.7 \cdot 10^{-10}]$, and $0.03175361/Z(59/50) \in [0.0085335161 \pm 5 \cdot 10^{-12}] > 0.0085$. A second run with $N = 20$, $N_G = 26$, 240 boxes, precision 192 passes with worst bound -0.03333053 (stability check, same code). [interval-certified]

Erratum on an earlier version (important). A first version of this certificate (Lean commit `1f3ace0`) was *invalid*: python-flint's default `ctx.cap` = 10 silently truncated every product of power series at κ^9, so the orders $\kappa^{10}, \dots, \kappa^{41}$ were neither computed nor bounded, and in each of the 160 boxes at least one exported bound (orders E_{12}–E_{40}) was below the true value. Lean accepted it, since Lean only sees the rational inequalities. The error was found by an independent adversarial re-verification, and its numerical effect is small ($\le 4.7 \cdot 10^{-6}$ against a margin 0.0317), but the artefact had to be regenerated. The repaired script makes cap and precision mandatory arguments, refuses cap $< L = 42$ (guard 1) and asserts that every series and product has length exactly L (guard 2, 851 840 assertions per run; the assertion is on `.prec` because `coeffs()` drops trailing zeros). The chordal certificate of §4.4 was written with the same guards from the start.

Independent checks (gates). All logs return code 0 unless a refusal is expected.

gate	outcome
G1: cap $\in \{42, 50, 100, 200\}$	identical results in all 160 boxes; cap = 10 refused by guard 1; cap = 10 and 41 with guard 1 disabled refused by guard 2
G2: explicit Cauchy convolution (no `arb_series`)	all 42 coefficients overlap; $\max \lvert\Delta\text{mid}\rvert = 1.5 \cdot 10^{-39}$
G3: $N \in \{16, 20, 24\}$ vs 32	$\lvert U_N - U_{32}\rvert$ nonzero, decreasing ($1.34 \cdot 10^{-10}$, $1.20 \cdot 10^{-12}$, $1.08 \cdot 10^{-14}$ at $\alpha = 0$, $\kappa = 1.18$), below ε_N
G4: $N_G \in \{21, 22, 26\}$	22 vs 26 overlap for $j \le 41$; with 21 nodes disagreement starts exactly at $j = 40$, as predicted by Gauss exactness
G5: Arb vs float polar grid	gap $\le 1.0 \cdot 10^{-8}$, decreasing as $O(h^2)$ under grid refinement
independent recomputation (Galerkin; code not included in the archive)	Galerkin solver at finite κ and exact E_2, E_4: consistent with every box bound; recomputed data identical byte for byte
independent check (exact rationals; `independent_check_geodesic/`)	$E_j(\alpha) \in \mathbb{Q}[\cos\alpha]$ computed exactly by a different method (no quadrature, no `arb_series`); all 3200 Lean coefficients of `7a211ed` (160 boxes $\times$ $E_2, \dots, E_{40}$) are true upper bounds, via rigorous box majorants with the invariant "majorant $\ge$ exact sample" checked everywhere; the same tool refutes the old commit `1f3ace0` (1634 false bounds, 160/160 boxes); worst `bP` $= -0.03175360$ in rationals; $0.0085335 \ge 0.0085$ checked exactly against a rigorous upper bound of $Z(K)$; U_{16} vs. an independent Galerkin solver at 700 points (κ, α): $\max \lvert\Delta\rvert = 1.34 \cdot 10^{-10} < \varepsilon_{16}$

Floating-point gates are [numerical]; they test the modelling, they are not part of the proof.

Lean. Commit `7a211ed0426e95b36a0ff35f5e896f8594b34fd3` of the author's Lean library (parent `1f3ace0`; only `KevYM/Certif/DobrushinLien/Donnees.lean` changed). The three modules are reproduced byte for byte in the standalone Lean project `lean/` of the accompanying archive (they are unchanged in `1b0f039`). Lean toolchain `v4.34.0`. Main statements (verbatim):

```
theorem sound (cs : List ℚ) (a b : ℚ) (ha : 0 ≤ a) (hneg : bP cs 0 a b < 0)
    (es : List ℝ) (hes : List.Forall₂ (fun (e : ℝ) (c : ℚ) => e ≤ (c : ℝ)) es cs)
    (w : ℝ) (haw : (a : ℝ) ≤ w) (hwb : w ≤ b) : pR es 0 w < 0
theorem certs_ok : certs.all (fun cs => decide (bP cs 0 0 KW < 0)) = true
theorem certificat_boites (cs : List ℚ) (hcs : cs ∈ certs) (es : List ℝ)
    (hes : List.Forall₂ (fun (e : ℝ) (c : ℚ) => e ≤ (c : ℝ)) es cs)
```

```
    (w : ℝ) (hw0 : 0 ≤ w) (hw : w ≤ (KW : ℝ)) : pR es 0 w < 0
theorem somme_dobrushin_lt_one (β : ℝ) (hβ : β < Real.pi / 16) :
    18 * β * (8 / (9 * Real.pi)) < 1
theorem kappa_max_lt : 6 * (Real.pi / 16) < (59 : ℝ) / 50
```

Here `bP` is the monotone upper bound $\sum_k \max(c_k a^k, c_k b^k)$, `pR es 0 w` $= \sum_j e_j w^j$ (`pR_eq_sum`), `KW` $= 3481/2500 = (59/50)^2$, and `certs` is the list of 160 exported rational coefficient vectors (Arb upper bounds rounded up to the grid 2^{-40}, re-checked exactly before export). `certs_ok` is proved by `decide +kernel`. The file also proves $18\cdot(\pi/16)\cdot 8/(9\pi) = 1$ and the converse for $\beta \geq \pi/16$. Build clean (0 warnings); all 13 theorems depend only on `propext`, `Classical.choice`, `Quot.sound`; the rebuild from `git archive 7a211ed` succeeds, and so does the from-scratch build of the standalone project (`lean/logs/build_from_zero.log`). [Lean-checked]

What Lean covers, exactly. Lean proves `bP cs 0 0 KW < 0` (strict) for each of the 160 exported vectors, and the implication of `certificat_boites`: *if* the true coefficients lie below the exported rationals, *then* $\Phi < 0$ on $[0, K^2]$; plus the threshold arithmetic. Not in Lean: the constant 0.0085 (i.e. $-\max \texttt{bP}/Z(K) \geq 0.0085$, checked exactly in rationals by the independent check against a rigorous upper bound of $Z(K)$) and the covering of $[0,\pi]$ by the boxes (the association box $\mapsto$ α-interval is a comment in the Lean file); both are [interval-certified].

Trust base. The implication "integrals $\Rightarrow$ exported bounds" rests on Arb and on the reduction of §5.1 (proved on paper; the classical inputs are Kantorovich–Rubinstein duality, quoted from [8, Thm 10.8], the Holley–Stroock perturbation argument, written out in Step 3, and the weak solvability of $-L\varphi = R$, written out in Step 1), and is *not* formalised. It is cross-checked in exact rationals by the independent check (table above), which shares with the certificate only the modelling (reduced ODEs, measure $ds\,d\tau$).

Artefacts (SHA-256). Paths are relative to the accompanying archive.

file	SHA-256
`certificate_geodesic/cert_lien.py` (v2)	`8b76b11125693b8f71b44898e959635066b3804560c2ee314092667a562ea096`
`certificate_geodesic/export_lean.py`	`1885f8bc2e8444bd616efa41c5c18bbfaa3fca78e6bc3e06ae197014160c1bcb`
`certificate_geodesic/cert_N16_G22_B160_cap200_p128.json`	`91700df0e8031e1f6ed4c3d6d9e88c42ddf36f0abeacd58c9f328738b0d9c43e`
`lean/KevYM/Certif/DobrushinLien/Donnees.lean` (= `7a211ed`)	`9d6c06283848a9ef38baecb222e195e9aa528009412e7c8bded6f2555321c37f`

6 Proofs of Theorems 1 and 2

Theorem 1. Let $\beta < 2/9$, and use the chordal metric. By Theorem 6, $\sup_{\kappa \leq 6\beta} D_{\mathrm{ch}}(\kappa) = D_{\mathrm{ch}}(0) = 1/4$ (as $6\beta < 4/3$, `kappa_max_lt`); by Proposition 5 each of the 18 neighbours has influence $\beta/4$ and all other links have influence 0; so (5.2) holds with $d = 9\beta/2 < 1$ (`somme_dobrushin_lt_one`). (5.1) holds for every field. By [1, Thm 4] there is at most one field with the Wilson single-link conditionals, hence at most one Gibbs measure (item 2 of §2); existence is Proposition 3. □

Theorem 2. Let $\beta < \pi/16$, and use the geodesic metric. By Theorem 10, $\sup_{\kappa \leq 6\beta} D(\kappa) = D(0) = 8/(9\pi)$, and Proposition 5 gives the sum $16\beta/\pi < 1$ (`somme_dobrushin_lt_one` of the geodesic module). For $\beta \geq \pi/16$ the lower bound of Proposition 5 gives a sum $\geq 16\beta/\pi \geq 1$. □

7 Relation to the literature

Bhattacharjee–Bhattacharya [17] (added in v1.4). After v1.3 of this note (28 September 2026, not publicly released) was completed, we learned of the public preprint [17], dated 18 October 2026. Its Theorem E (p. 2) reads (text of the PDF): "Let $G = SU(2)$ and $|\beta| < 1/(6(d-1))$. The Wilson theory on $\mathbb{Z}^d$ has exactly one infinite-volume Gibbs measure, and its truncated correlations of local Lipschitz functions decay exponentially with every rate $\lambda < -\log 6(d-1)|\beta|$. At $d = 4$ the range is $\beta_W < 2/9$. No single-link Wasserstein–Dobrushin argument, in the chordal or the geodesic metric, reaches beyond this range." with $\beta_W = N^2\beta$ (their footnote 4, the normalisation of [2]); its Theorem F adds a spectral gap of the transfer matrix in this range, and its Theorem 7 (p. 9) states that for every $c \in \mathbb{R}^4$ and unit $h \in \mathbb{R}^4$, "$L(c,h) \le \frac{1}{4}$, with equality at $c = 0$. Equivalently, $W_1(\nu_b, \nu_{b'}) \le \frac{1}{4}|b - b'|$ for all $b, b' \in \mathbb{R}^4$, and the constant $\frac{1}{4}$ is the best possible." [read]. To our knowledge [17] is the first public statement of the range of our Theorem 1, with stronger conclusions, and of the maximum-at-zero-field property of Levin's Remark 3.7 for $N = 4$ and *all* fields. We make *no priority claim*, and we have not checked the proofs of [17] in full. What this note provides independently is: the exact one-link coefficient $D_{\mathrm{ch}}(\kappa) = \mathbb{E}_\kappa[x_\perp^2]$ for $\kappa \le 4/3$, with the explicit remainder of Theorem 6; the exact geodesic coefficient $8\beta/(9\pi)$ for $\kappa \le 3\pi/8$ (Theorem 10) and the range $\pi/16$; two independent interval-arithmetic certificates; the Lean-checked final steps of §4.4 and §5.2; and the transfers of the published constants of Levin, Royer, Gross, Simon and Shen–Zhu–Zhu to the Wilson link (Table 2). A later version of the same preprint treats a different object (limits with free boundary conditions, gauge-invariant observables), which we do not compare with Theorem 1. Nissim [18] treats $U(N)$ at sufficiently small 't Hooft coupling; no numerical $SU(2)$ threshold is stated there.

Shen–Zhu–Zhu [2]. Their action is (verbatim LaTeX from the arXiv source, eq. (1.2), p. 4 of the arXiv PDF)

```
\cS (Q) \eqdef N\beta  \Re \sum_{p\in \CP^+_{\Lambda_{L}}} \Tr(Q_p),
```

i.e. $\mathcal{S}(Q) = N\beta \operatorname{Re} \sum_{p \in \mathcal{P}^+} \operatorname{Tr}(Q_p)$, one term per plaquette; their Assumption 1.1 is equivalent to (1.3), p. 4: $|\beta| < \frac{1}{16(d-1)}$ for $G = SU(N)$. For $SU(2)$, $2\beta_{\mathrm{tH}} \operatorname{Re}\operatorname{Tr} = 4\beta_{\mathrm{tH}} \cdot \frac{1}{2} \operatorname{Re}\operatorname{Tr}$, so $\beta_W = 4\beta_{\mathrm{tH}}$ and their range is $\beta_W < 1/12 \approx 0.0833$ in $d = 4$. Under it they prove (Theorem 1.2, p. 5) uniqueness of the invariant measure of the Langevin dynamics and convergence of all tight limits of the finite-volume measures, together with a log-Sobolev inequality (Theorem 1.4) and exponential decay of correlations (Corollary 1.6). On p. 6 (arXiv v1, the only arXiv version; the published text was not compared) they write (verbatim): "We remark that uniqueness for small β could possibly also be proven using the method of Dobrushin, see e.g. [Dob70]. To this end one would also need to consider the related Wasserstein metric with respect to the Riemannian distance similarly as we do in this paper. However as we understand such an argument has not been carried out in detail for lattice Yang Mills in the literature." The present note carries out this programme for the gauge model, for uniqueness only (with the Riemannian distance in Theorem 2, and with the Euclidean distance of $\mathbb{R}^4$ in Theorem 1). The single-site ingredient (W_1 on a sphere, derivative along a path of fields written as a covariance) is not new: it is the method of Levin [11] (Euclidean metric) and Royer [10] (geodesic metric) for rotator models, and Levin's many-body version already covers the Wilson model (next paragraphs). Neither paper mentions gauge theories. Our uniqueness statement is directly of DLR type; we do *not* claim that [2] lacks DLR uniqueness (their statement is formulated for the Langevin invariant measure and limits with periodic, Dirichlet "or other" boundary conditions).

Levin: the Euclidean W_1 criterion on spheres, many-body interactions. Levin [11] (read on the page images of the Project Euclid scan) considers "the lattice model on $\mathbb{Z}^\nu$ with

single spin space S^{N-1}, $N \geqslant 2$" with "a priori measure μ_0 the invariant probability measure on S^{N-1}" (p. 65) and the "Vaserstein distance" $V(\mu,\nu) = \sup_{\|f\|_L=1} |\mu(f) - \nu(f)|$ over functions of Lipschitz norm 1 for "$\|\cdot\|_2$ [...] Euclidean norm on $\mathbb{R}^N$" ((3.3)–(3.4), p. 67), i.e. the *chordal* metric of $S^{N-1} \subset \mathbb{R}^N$. His Theorem 3 (p. 67) reads: "For $\mathbf{h}, \mathbf{h}' \in \mathbb{R}^N$, let $\mu_{\mathbf{h}}, \mu_{\mathbf{h}'}$ be probability measures on S^{N-1} given by $d\mu_{\mathbf{h}}(\mathbf{x}) = \frac{1}{Z_{\mathbf{h}}} e^{\mathbf{h}\cdot\mathbf{x}} d\mu_0(\mathbf{x})$ [...] Then, $V(\mu_{\mathbf{h}}, \mu_{\mathbf{h}'}) \leqslant \frac{\sqrt{5}}{N}\|\mathbf{h}' - \mathbf{h}\|_2$. (3.8)", proved by writing the distance "as the variation of an expectation along an appropriate path in **h**-space" (p. 68), with two lemmas controlling "separately variation parallel to the 'magnetic field', and perpendicular to the 'magnetic field'," the perpendicular one being Lemma 3.2 (p. 68, quoted in §4.1). Remark 3.6 (p. 73) extends the uniqueness theorem to "many-body interactions of the form $\sum J_X^{(i_a)} \prod_{a\in X} \sigma_a^{i_a}$" with the condition "$T > T_C^N \equiv (\sup_a \sum_{X\ni a}(|X|-1)\mathscr{J}_X)\frac{\sqrt{5}}{N}$, (3.49)", where $\mathscr{J}_X$ is the supremum of the interaction over configurations; the details are "[left] to the reader". Remark 3.7 (p. 73) conjectures that "the numerical factor of $\sqrt{5}$ appearing in Theorem 1 can be replaced by 1" and notes that "the conjecture is true if one shows $\sup_{\|f\|_L=1} |\mu_{\mathbf{g}}[(\mathbf{x}\cdot\hat{\mathbf{c}} - \langle \mathbf{x}\cdot\hat{\mathbf{c}}\rangle_{\mathbf{g}})f]|$ [...] attains its maximum value at $\mathbf{g} = 0$". [read]. The paper does not mention gauge theories.

Transfer to the Wilson link (our computation from the published statement, not a published result). As noted in §3, the Wilson weight $e^{(\beta/2)\operatorname{Re}\operatorname{Tr} U_p} = e^{\beta\langle u_1 u_2 \bar{u}_3 \bar{u}_4, \mathbf{1}\rangle}$ is of Levin's form (3.48) with $N = 4$, $|X| = 4$ and $\mathscr{J}_X = \beta$; each link lies in 6 plaquettes, so (3.49) at unit temperature reads $6 \cdot 3 \cdot \beta \cdot \sqrt{5}/4 < 1$, i.e. $\beta_W < 2\sqrt{5}/45 \approx 0.09938$. [proved] from [read] (exact arithmetic; log `calc/levin_transfer.log`). This already exceeds $1/12$.

What Theorem 6 adds. The supremum in Levin's Remark 3.7 is our $D_{\mathrm{ch}}(|\mathbf{g}|)$ for $N = 4$. Theorem 6 shows that it attains its maximum at $\mathbf{g} = 0$ among $|\mathbf{g}| \leq 4/3$, with value $1/N = 1/4$, and identifies it exactly ($\mathbb{E}_{\mathbf{g}}[x_\perp^2]$, attained by the coordinate orthogonal to the field). It therefore confirms, for $N = 4$ and $|\mathbf{g}| \leq 4/3$, the maximum-at-$\mathbf{g} = 0$ property conjectured in [11, Rem. 3.7]; it does not prove the conjecture, which is stated for all N, all $\mathbf{g}$ and general interactions. Since the fields that occur for $\beta < 2/9$ satisfy $|\mathbf{g}| \leq 6\beta < 4/3$, the Wilson threshold is Levin's transferred bound with $\sqrt{5}$ replaced by 1: $2\sqrt{5}/45 \cdot \sqrt{5} = 2/9$ exactly.

Royer: the geodesic W_1 criterion on spheres. Royer [10] (read on the Numdam page images) works with "the unit sphere S^n with its natural (geodesic) distance r" and Dobrushin's distance "$\mathrm{R}(\nu_1,\nu_2) = \sup\{\int f\,d\nu_1 - \int f\,d\nu_2;\ \mathscr{L}(f) \leqslant 1\}$" (p. 30). He defines $C(n,h)$ as "the best constant in the inequality $\operatorname{var}(f,\nu(h)) \leqslant C(n,h)(\mathscr{L}(f))^2$" and $C(n) = \sup_h C(n,h)$ (Lemma 1.2, p. 31), proves (3)–(4) quoted in §3, and bounds $C(n)$: the proof of Corollary 2.3 (p. 34) gives "$C(n,h) \leqslant \chi(n,h) < (n-2)^{-1}$; thus $C(n) \leqslant (n-2)^{-1}$", Corollary 3.5 (p. 37) "$C(n,0) = v_n$", and Theorem 3.6 (p. 37) "For $n > 1$, $C(n) < v_n + v_{n-1}$", with $v_n + v_{n-1} = 2\sum_{k\geqslant n} k^{-2}$ (p. 37). He states that his estimates "improve on that given by Levin [10], namely $N/\sqrt{5}$, for $N \geqslant 3$" (p. 30), and on p. 38 he conjectures the geodesic analogue of Levin's Remark 3.7: "one might think that $C(n,h) \leqslant C(n,0)$". [read].

Transfer to the Wilson link (our computation from the published statements, not a published result). By (3) with $n = 3$, $K = \beta$ and z the staple moved by the neighbour, each neighbour has influence at most $\beta(C(3)/4)^{1/2}$, hence uniqueness when $18\beta(C(3)/4)^{1/2} < 1$. With $C(3) \leq 1$ (Cor. 2.3) this gives $\beta_W < 1/9 \approx 0.1111$; with $C(3) < v_3 + v_2 = \pi^2/3 - 5/2$ (Thm 3.6) it gives $\beta_W < \left(9\sqrt{\pi^2/3 - 5/2}\right)^{-1} \approx 0.12502$. [proved] (exact arithmetic; SymPy logs `calc/royer_transfer.log` and `calc/chiffres_v1.log`). At zero field, Royer's route gives $\beta(v_3/4)^{1/2}$ with $(v_3/4)^{1/2} = \frac{1}{2}(\pi^2/12 - \frac{1}{2})^{1/2} \approx 0.28393$, only 0.35% above our exact $D(0) = 8/(9\pi) \approx 0.28294$; but this is not uniform in the field and gives no uniqueness statement.

Where the gain comes from. Our main threshold $2/9$ exceeds the transferred Royer bound 0.12502 by a factor 1.777, Levin's transferred bound by exactly $\sqrt{5} \approx 2.236$, and $1/12$ by $8/3$; the geodesic threshold $\pi/16$ exceeds 0.12502 by 1.5705, and $2/9$ exceeds $\pi/16$ by 13.2%. The

gain is not in the constants at $h = 0$: it comes from computing the exact single-link coefficient as a function of $\kappa = \beta|s|$ and showing that it is maximal at $\kappa = 0$ on $[0, 6\beta]$ (Theorems 6 and 10), which replaces the field-uniform constants $\sqrt{5}/N$ (Levin) and $C(n)$ (Royer). This maximality is the property conjectured by Levin (Euclidean metric, all N, all fields) and, for the variance constant, by Royer; we establish it only for $N = 4$ and the fields that occur, for the Euclidean W_1 coefficient ($\kappa \le 4/3$) and for the geodesic one ($\kappa \le 3\pi/8$), by computer-assisted proofs. For $N = 4$ the Euclidean property is proved for all fields in [17, Theorem 7]; we did not check whether [17] also covers the geodesic coefficient of Theorem 10, and we found no other published proof of either property (literature search not exhaustive).

Remark 13 (Numerical evidence and quasi-optimality of the field). Before Theorem 6 was proved, linear programs for $\sup_f$ (reduction to functions of two coordinates, HiGHS; logs `calc/lp2d_m14.log` and `calc/check1d.log`) gave for $\max_\alpha D_{\mathrm{ch}}(\kappa, \alpha)$ the values 0.2500, 0.2474, 0.2402, 0.2332, 0.2164 at $\kappa = 0$, 0.5, 1, 4/3, 2, the maximum over directions being attained for $v \perp \hat{s}$; the same code reproduces the geodesic $D(0)$ to 0.15% [numerical]. These values agree with $\mathbb{E}_\kappa[x_2^2] = 0.25$, 0.24744, 0.24019, 0.23332, 0.21656 (the discretised programs are neither upper nor lower bounds; the value $0.2164 < 0.21656$ at $\kappa = 2$ is a discretisation error, since $\mathbb{E}_\kappa[x_2^2]$ is an exact lower bound for $D_{\mathrm{ch}}(\kappa)$). The full flux bound of the explicit field (4) is within $1.7 \cdot 10^{-4}$ of the linear programs for all directions tested, so the field is close to optimal [numerical]. Beyond $\kappa = 4/3$ nothing is certified; the ratio $\max(J_0, J_1)/(\mathbb{E}x_2^2 - \operatorname{Var} x_1)$ grows from 0.32 at $\kappa = 4/3$ to 0.47 at $\kappa = 2$ [numerical].

Total variation. The classical total-variation form of Dobrushin's criterion for many-body interactions on a compact single-spin space is due to Gross [13] and Simon [14], neither of whom mentions gauge theories. Gross [13] (read on the page images) treats "X a compact metric space", with "the cases of principle current interest [...] X = finite set, S^m, compact group, or a finite product of these" (p. 10), coefficients $\varrho_{a,b} = \frac{1}{2}\sup\|\mu_b(\cdot|s) - \mu_b(\cdot|t)\|_{\mathrm{Var}}$ ((2.3), p. 11), and proves (Theorem 3, p. 22): "If $\alpha_0 < 1$ then φ has a unique Gibbs state σ" with the decay bound (4.9), where $\alpha_0 = 2e^{4\|\varphi\|_1}\sup_a \sum_{b\neq a}\zeta(a,b)e^{d(a,b)}$ (4.8), and "$\|\mu_a(\cdot|t) - \mu_a(\cdot|s)\|_{\mathrm{Var}} \leqslant e^{4\zeta(a,b)} - 1$" ((4.11), Lemma 4.1). [read]. Simon [14] (read on the page images) proves: "If $\sum_{0\in X}(|X|-1)\|\Phi(X)\|_\infty < 1$, then (2) holds" (Theorem, p. 184), with "$\|\Phi(X)\|_\infty$ [replaceable] by $\frac{1}{2}[\max(\Phi(X)) - \min(\Phi(X))]$" (Remark 1, p. 185), via the Lemma "$\|\mu_h - \mu_g\| \leqslant \|h - g\|_\infty$" whose proof writes $\frac{d}{d\theta}\nu_\theta(f)$ as a covariance ((3)–(4), p. 184); in Remark 5 (p. 185) he observes that a one-site quantity "takes its maximum at $a = 0$ by the GHS inequality" for spins in $[-1, 1]$. [read]. *Transfer to the Wilson link* (ours): with $\Phi(p) = -\frac{\beta}{2}\operatorname{Re}\operatorname{Tr} U_p$, Simon's condition with his Remark 1 reads $18\beta < 1$, i.e. $\beta_W < 1/18 \approx 0.056$; Gross's Theorem 3 gives $36\beta e^{24\beta} < 1$, i.e. $\beta_W < 0.0180$, and his (4.11) in his Theorem 1 gives $\beta_W < \ln(10/9)/4 \approx 0.0263$ [proved] from [read] (log `calc/gross_transfer.log`). Chatterjee [12] (read in arXiv:1811.09770v3) applies the total-variation criterion to $\mathbb{Z}_2$ (Ising) lattice gauge theory in $d = 4$: "The easiest way to prove exponential decay of correlations at strong coupling is by using Dobrushin's condition" (p. 21) and, in the proof of Theorem 5.2 (p. 24), "There are at most 18 such neighbors", with a total-variation bound $L_0\lambda$ per neighbour, L_0 a universal constant that is not made explicit. [read]. For the $SU(2)$ single-link conditionals the exact one-link total-variation coefficient is $c_{\mathrm{TV}}(\beta) = \sup\|q_s - q_{s'}\|_{\mathrm{TV}} = \frac{4\beta}{3\pi} + O(\beta^2)$, the supremum being over $|s|, |s'| \le 6$ differing by one staple, so $|s - s'| \le 2$. [proved]: *upper bound* (this is the route (3) of Simon's Lemma, without the Schwarz step), along the segment $s(t) = s + t\delta$, $\delta = s' - s$, $|s(t)| \le 6$, the density h_t of $q_{s(t)}$ w.r.t. Haar measure satisfies $e^{-12\beta} \le h_t \le e^{12\beta}$, and $\frac{d}{dt}q_{s(t)} = \beta(\langle u, \delta\rangle - \mathbb{E}_{q_{s(t)}}\langle u, \delta\rangle)q_{s(t)}$, so $\|q_s - q_{s'}\|_{\mathrm{TV}} \le \frac{\beta}{2}\sup_t\left(\mathbb{E}_{q_{s(t)}}|\langle u, \delta\rangle| + |\mathbb{E}_{q_{s(t)}}\langle u, \delta\rangle|\right) \le \beta(e^{12\beta}\frac{4}{3\pi} + e^{12\beta} - 1)$, using $\mathbb{E}_{\mathrm{Haar}}|\langle u, e\rangle| = 4/(3\pi)$ and $\mathbb{E}_{\mathrm{Haar}}\langle u, \delta\rangle = 0$; *lower bound*, take $s = 0$ and $s' = -2w$ (flip the staple $w \mapsto -w$, the other five summing to $-w$), then $\|q_0 - q_{s'}\|_{\mathrm{TV}} = \frac{1}{2}\mathbb{E}_{\mathrm{Haar}}|e^{\beta\langle u, s'\rangle}/Z - 1| = \frac{\beta}{2}\mathbb{E}_{\mathrm{Haar}}|\langle u, s'\rangle| + O(\beta^2) = \frac{4\beta}{3\pi} + O(\beta^2)$. The resulting sum $24\beta/\pi$ gives the threshold $\approx \pi/24 \approx 0.131$ *at leading order only*;

the finite-β threshold, from a floating-point evaluation of $c_{\mathrm{TV}}(\beta)$, is ≈ 0.1310 [numerical]. The ratio of the first-order coefficients is $\frac{12/(9\pi)}{8/(9\pi)} = \frac{3}{2}$ for the geodesic metric, and $\frac{4/(3\pi)}{1/4} = \frac{16}{3\pi} \approx 1.70$ for the chordal one; the "gain of W_1 over total variation" therefore depends on the metric.

Summary: five levels. (i) Total variation: transferred published constants give $\beta_W < 1/18$ (Simon) [proved] from [read]; with the exact one-link coefficient, $\beta_W \approx \pi/24 \approx 0.131$ (leading order [proved], value [numerical]). (ii) Euclidean W_1 with Levin's published constant transferred to the Wilson link: $\beta_W < 0.09938$ [proved] from [read]. (iii) Geodesic W_1 with Royer's published constants transferred: $\beta_W < 0.12502$ (Thm 3.6) or $< 1/9$ (Cor. 2.3) [proved] from [read]. (iv) Geodesic W_1 with the exact coefficient, uniform in $|h| \le 6\beta$ (Theorem 2): $\beta_W < \pi/16 \approx 0.19635$. (v) Euclidean W_1 with the exact coefficient, uniform in $|h| \le 6\beta$ (Theorem 1): $\beta_W < 2/9 \approx 0.2222$. Levels (ii)–(v) exceed $1/12$ rigorously; the exact-coefficient TV level does so only at leading order and numerically, since the explicit upper bound on c_{TV} written above proves the total-variation criterion only for $\beta < 0.0414$.

Bakry–Émery cannot pass $\beta_W = 1/8$.

Proposition 14. *Equip each factor $SU(2)$ with a bi-invariant metric (any constant scale per link). At the π-flux configuration ($U_\ell = \pm\mathbf{1}$ with every $U_p = -\mathbf{1}$; it exists on $\mathbb{Z}^4$ and on periodic boxes of even side) the quadratic form $\mathrm{Ric} + \beta \,\mathrm{Hess}\, S_W$, $S_W = \sum_p (1 - \frac{1}{2}\,\mathrm{Re}\,\mathrm{Tr}\, U_p)$, is not positive definite for $\beta > 1/8$ (and, on even periodic boxes, has the eigenvalue 0 at $\beta = 1/8$, so it is not positive definite there either). Hence the global Bakry–Émery condition $\mathrm{Ric} + \beta \,\mathrm{Hess}\, S_W \ge K > 0$ on link variables fails for $\beta_W \ge 1/8$.* [proved]

Proof. In the Frobenius normalisation ($X_a = i\sigma_a/\sqrt{2}$), $\exp(sX_a)$ has rotation angle $s/\sqrt{2}$, so $SU(2)$ is the round S^3 of radius $\sqrt{2}$ and $\mathrm{Ric} = g$. Both Ric and $\mathrm{Hess}\, S_W$ are $(0,2)$-tensors unchanged by constant rescalings of the factors, so the sign question is scale-free. Perturb $U_\ell \mapsto e^{X_\ell} U_\ell$. At a configuration with all U_ℓ central ($\pm\mathbf{1}$), $U_p \mapsto U_p\, e^{X_1} e^{X_2} e^{-X_3} e^{-X_4}$, so the second-order expansion of $1 - \frac{1}{2}\,\mathrm{Re}\,\mathrm{Tr}\, U_p$ at $U_p = -\mathbf{1}$ is minus the one at $U_p = \mathbf{1}$, and both configurations are critical. Therefore $\mathrm{Hess}\, S_W(\pi\text{-flux}) = -\mathrm{Hess}\, S_W(\text{flat}) = -\frac{1}{2}\, d^*d \otimes I_3$, where $d^*d = B^{\mathsf{T}}B$ with B the lattice curl (since $1 - \frac{1}{2}\,\mathrm{Re}\,\mathrm{Tr}\, e^Y = |Y|^2/4 + O(|Y|^3)$ and the commutator terms of Y are traceless). By Fourier analysis the largest eigenvalue of d^*d is $4d = 16$, attained at momentum (π,π,π,π) with $d - 1 = 3$ transverse polarisations. So $\lambda_{\min}(\mathrm{Hess}\, S_W) = -8$ with multiplicity 9 on even periodic boxes (on $\mathbb{Z}^4$: infimum of the quadratic form, via Weyl sequences), and $\mathrm{Ric} + \beta \,\mathrm{Hess}\, S_W$ has the eigenvalue $1 - 8\beta$. □

Checks [numerical]: an independent arbiter script (finite differences, $L = 2$) finds $\lambda_{\min} = -8.000000$, multiplicity 9, $\max |H(\pi) + H(\text{flat})| = 7.5 \cdot 10^{-8}$; a second script written for this note finds $\max \mathrm{eig}(B^{\mathsf{T}}B) = 16$ with multiplicity 3 on $L = 2$ and $L = 4$, and $Q_\pi(v) = -Q_{\text{flat}}(v) = -\frac{1}{2} v^{\mathsf{T}}(B^{\mathsf{T}}B \otimes I_3)v$ to $2 \cdot 10^{-7}$ along random directions on $L = 4$. Whether $\sup_U(-\lambda_{\min}\mathrm{Hess}\, S_W) = 8$ (so that global Bakry–Émery holds up to exactly $1/8$) is [conjecture]. Consistently, [2] bound |Hess| by $12\beta_W$ and obtain $1/12$. Since $2/9 > \pi/16 > 0.12502 > 1/8$ (the transferred Royer bound exceeds $1/8$ by a margin of $2 \cdot 10^{-5}$; and, at leading order only, $\pi/24 > 1/8$), Dobrushin-type criteria, which need no convexity, go beyond the ceiling of the global Bakry–Émery route on link variables with bi-invariant product metrics. This does not contradict [5], which obtains an area law up to $\beta_W < 1/6$ by applying Bakry–Émery to a different (vertex) spin system.

Other works. Osterwalder–Seiler [3] prove, by cluster expansion, decay, a mass gap and uniqueness of the infinite-volume limit for "g_0 sufficiently large", without an explicit constant [read]. Chatterjee [6] (arXiv:2006.16229v5, p. 7, after Definition 2.3; the published version was not compared) states that exponential decay under arbitrary boundary conditions "can

be shown to hold for essentially any lattice gauge theory when β is small enough, using the methods of Dobrushin and Shlosman" [7], without a constant [read]. Cao–Nissim–Sheffield [5] (arXiv, not yet refereed) prove an area law on the torus for $\beta_{\mathrm{tH}} < \frac{1}{8(d-1)}$, i.e. $\beta_W < 1/6$; this is a different statement and is not compared with ours. Seiler's monograph [4] could not be obtained and was *not consulted*; for Georgii's book [16] only the table of contents was read (Chapter 8 "Uniqueness", §8.1 "Dobrushin's condition of weak dependence", p. 140), and its content was *not consulted*; Simon's book [15] was *not consulted*. No claim of this note rests on any of them (see §8, item 4).

source	statement	range in β_W ($SU(2)$, $d = 4$)	status
[3]	decay, gap, unique limit	small, not explicit	[read]
[12]	TV-Dobrushin, decay, $\mathbb{Z}_2$ gauge, $d = 4$	not explicit (L_0)	[read]; other group
[13] Thm 3, transferred (here)	uniqueness, decay	< 0.0180	[proved] from [read]
[14] Thm + Rem. 1, transferred (here)	uniqueness	$< 1/18 \approx 0.056$	[proved] from [read]
[2]	uniqueness, LSI, decay	$< 1/12 \approx 0.083$	[read]
[11] Rem. 3.6, transferred (here)	uniqueness	$< 2\sqrt{5}/45 \approx 0.0994$	[proved] from [read]
[10] Cor. 2.3, transferred (here)	uniqueness	$< 1/9 \approx 0.111$	[proved] from [read]
Prop. 14	global BE on links fails	$\geq 1/8 = 0.125$	[proved]
[10] Thm 3.6, transferred (here)	uniqueness	< 0.12502	[proved] from [read]
TV-Dobrushin, exact one-link coefficient (here)	uniqueness	$\approx \pi/24 \approx 0.131$	leading order [proved], value [numerical]
[5]	area law on torus (different)	$< 1/6 \approx 0.167$	[read], preprint
Theorem 2 (geodesic W_1, exact coefficient)	uniqueness (DLR) only	$< \pi/16 \approx 0.196$	see Thm 2
Theorem 1 (Euclidean W_1, exact coefficient)	uniqueness (DLR) only	$< 2/9 \approx 0.222$	see Thm 1
[17] Thms E, F (preprint, 18 Oct. 2026)	uniqueness, decay, transfer-matrix gap	$< 2/9 \approx 0.222$	[read]; proofs not checked here

Table 2: Explicit ranges, all converted to β_W for $SU(2)$ in $d = 4$ (Table 1). Rows marked "transferred (here)" apply published statements of [13, 14, 11, 10] to the Wilson link; they are not published results, and none of these papers mentions gauge theories.

8 Limitations

1. *Uniqueness only.* No decay of correlations, spectral gap, log-Sobolev inequality or area law is claimed at $\beta < 2/9$ in this note. The preprint [17] states exponential decay and a transfer-matrix gap in the same range (proofs not checked here); among refereed publications, the best explicit range we know for these conclusions remains 1/12 [2]. Explicit decay statements covering four-body plaquette interactions do exist in total variation: Gross's Theorem 3 [13], transferred, gives uniqueness and exponential decay for $\beta_W < 0.0180$, far below 1/12. Gross's Theorem 1 [13, p. 11] turns any total-variation Dobrushin sum below 1 into exponential decay (take $d(a, b) = r|a - b|$ with $r > 0$ small); with a certified bound on c_{TV} it would give decay up to the TV threshold ≈ 0.131, hence

beyond 1/12 [conjecture] (not certified, not written out). For the W_1 criterion, Levin [11, Remark 3.5, p. 72] states that, "parallel to the discussion of Gross", the truncated spin–spin correlations decay exponentially under the W_1 condition with a semi-metric weight (his (3.46)–(3.47)); applied to our influence matrices this would plausibly give exponential decay of correlations of Lipschitz observables for $\beta < 2/9$ [conjecture]. Nothing is claimed.

2. *Formalisation boundary.* For the chordal certificate, the flux bound, the explicit field and the elimination of the direction (§§4.2–4.3) are proved on paper; the box bounds, the positivity of m and the constant 0.0084 are interval-certified; Lean checks only the final rational implication, two algebraic steps and the threshold arithmetic. For the geodesic certificate, the chain "integrals → coefficient bounds" (Arb, reduction of §5.1) is outside Lean, and so are the constant 0.0085 and the covering of $[0, \pi]$ by the boxes (§5.2). Kantorovich–Rubinstein duality is quoted from [8, Thm 10.8, p. 53]; McShane's extension, the divergence theorem for Lipschitz fields, the Holley–Stroock perturbation argument [9] and the weak solvability of $-L\varphi = R$ (Lax–Milgram + Poincaré) are standard and written out here; none of them is formalised.

3. *History and audit of the certificates.* The first geodesic certificate was wrong (see the erratum); the present one passed guards designed to catch that failure mode and two independent recomputations. The chordal certificate was written with the same guards and is confirmed by an independent implementation with different arithmetic, tail bounds and quadrature (§4.4); its mathematical reduction was re-derived line by line in that independent verification. Neither certificate, and none of the proofs, has been audited or refereed by a party outside the project.

4. *Novelty and prior art.* The single-site W_1 machinery on spheres is due to Levin [11] (Euclidean metric, many-body multilinear interactions, which cover the Wilson model; transferred, 0.0994) and Royer [10] (geodesic metric; transferred, 0.12502). The field-uniformity of the one-site coefficient (maximum at zero field) was conjectured by both, and the total-variation criterion for many-body interactions is due to Gross [13] and Simon [14]. None of these papers mentions gauge theories, and none contains $\beta < \pi/16$, $\beta < 2/9$ or a better threshold. The contribution of this note is the exact field-uniform single-link coefficient for the Euclidean and for the geodesic metric, its certification, and the resulting thresholds $2/9$ and $\pi/16$. *Limitation of the literature search:* Seiler's monograph [4] (not obtainable), the content of Georgii's book [16] (table of contents only) and Simon's book [15] were not consulted; any of them may contain a Dobrushin-type statement for plaquette interactions or for $SU(2)$ gauge fields. No claim of this note rests on them, but a priority claim for the gauge application should be read with this limitation in mind.

5. *Range of fields.* The chordal identity is certified only for $\kappa \le 4/3$, which is exactly what $\beta < 2/9$ requires; whether $D_{\mathrm{ch}}(\kappa) \le 1/4$ for all κ (Levin's conjecture for $N = 4$) is not addressed. Likewise the geodesic bound is certified only for $\kappa \le 59/50$.

6. *The value* $\pi/24$ is exact only at leading order in β; the finite-β TV threshold is a floating-point estimate, and has not been recertified in interval arithmetic.

7. *Physics.* The result is a strong-coupling statement ($g^2 > 18$, with $\beta_W = 4/g^2$). It says nothing about the continuum limit or the weak-coupling regime relevant to the Yang–Mills mass gap problem.

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